<u>Home Paper Final Submission</u>

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Design of a plant to produce 23,000 TPA of 2-hydroxyethyl methacrylate

by

NIDHI KULKARNI

22CHE142

Institute of Chemical Technology, Mumbai

(University under Section 3 of UGC Act 1956;

Elite Status and Centre of Excellence, Government of Maharashtra)

Maharashtra, India

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Chapter 1: Introduction

2-Hydroxyethyl Methacrylate, abbreviated as **HEMA**, is a hydroxy ester of methacrylic acid. It is a monomer that is used to make hydrogels used in making UV cured coatings and adhesives. The polymer (poly Hydroxyethyl Methacrylate or **PHEMA**) is used to make hydrogels and contact lenses and are used for drug delivery in biomedical fields. Since the monomer polymerizes quite easily, hydroquinone or methoxyphenol are added as inhibitors.

1.1 Product definition and properties

Name: 2-Hydroxyethyl Methacrylate (HEMA)

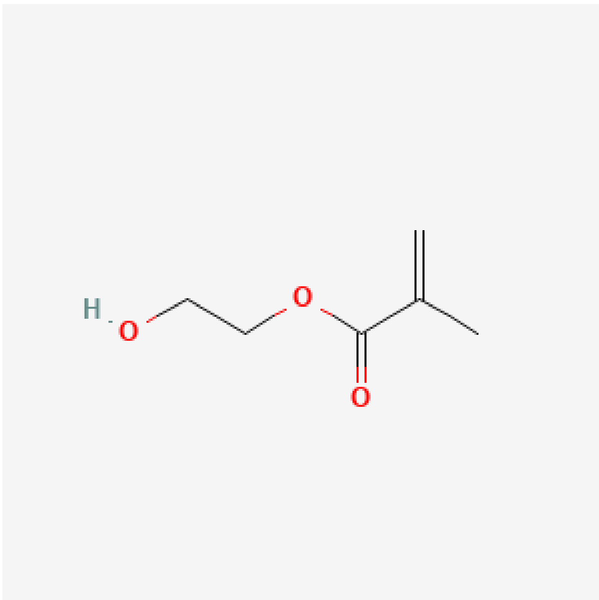


Figure 1: Structure of 2-Hydroxyethyl Methacrylate

Molecular formula: C6H10O3

Appearance: Colourless, viscous liquid

CAS No: 868-77-9

IUPAC name: 2,2-hydroxy,ethyl-methacrylate

Other names: 2-hydroxyethyl 2-methylprop-2-enoate, Glycol methacrylate, and ethylene glycol methacrylate, Glycol mono-methacrylate

Table 1: Physico-chemical Properties of HEMA

Property	**Value**
Molecular Weight	130.14 g/mol
Density (@ 20°C)	1.07 g/cm^3
Melting point	-12 ℃
Boiling point (@1 atm)	205 ℃
Flash point (@1 atm)	107 ℃
Solubility in water	Completely miscible
Vapor pressure (@ 20°C)	20 Pa

1.2

Hazards and Toxicity (Sigma-Aldrich, 2006)

Signal word: Warning.

Hazard statements: Causes skin irritation. May cause an allergic skin reaction. Causes serious eye irritation.

Precautionary statements: Prevention: P261 Avoid breathing mist or vapors. P264 Wash skin thoroughly after handling. P280 Wear protective gloves/ eye protection/ face protection. Response: P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention. P337 + P313 If eye irritation persists: Get medical advice/ attention. P362 + P364 Take off contaminated clothing and wash it before re-using.

1.3 Uses of HEMA

The industrial and biomedical applications of HEMA (2-hydroxyethyl methacrylate) encompass a broad spectrum due to its versatile chemistry and properties. HEMA’s high hydrophilicity and polymerizability make it indispensable in developing materials where mechanical strength, biocompatibility, and aqueous interaction are critical.

1.3.1 Industrial Applications

1. Synthesis of coating resins for automotive, architectural, and paper industries
2. Production of advanced adhesives and sealants
3. Textile finishing agents to improve fabric durability and water resistance
4. Printing inks for high-performance applications
5. Manufacture of hygiene products and plastic modifiers
6. Components in UV-curable and photopolymerizable formulations

1.3.2 Biomedical Applications

1. Contact lens hydrogels due to excellent water absorption and biocompatibility
2. Biosensors and membranes leveraging its inertness and hydrogel formation
3. Dental restorative materials such as adhesives and composite resins for hydrolytic and enzymatic stability
4. Controlled drug delivery systems using hydrogel networks for sustained release
5. Artificial organs and tissue engineering scaffolds

6. Cell culture substrates facilitating spheroid formation in biotechnology research

1.4 Market Analysis

Global production of 2-Hydroxyethyl Methacrylate (HEMA) is estimated to be around 300,000–350,000 tonnes per annum. The global market size in 2024 is valued at approximately USD 1.2 billion, with a compound annual growth rate (CAGR) of 6.5% for the forecast period 2024–2030. HEMA demand is primarily driven by its use in adhesives, coatings, resins, UV-curable materials, and contact-lens formulations.

According to trade statistics and industry reports, India currently has no domestic manufacturers of HEMA, and the country is entirely import-dependent. Imports during 2023–2024 are estimated at 7,000–8,000 tonnes per annum, mainly sourced from China, Japan, and Europe. Based on average import prices ranging between USD 2.5–10.5 per kg, the annual import value is estimated between USD 18 million and USD 84 million. There is negligible export activity from India, as all imported volumes are consumed domestically for downstream resin and coating applications. The exports are a very negligible quantity.

Globally, the largest manufacturer is Wanhua Chemical Group, operating a plant with a capacity of 39,500 tonnes per annum in China. Other significant producers include Dow Chemical Company (USA) and Mitsubishi Chemical Corporation (Japan). Smaller contributions come from Evonik Industries, Nippon Shokubai, and GEO Specialty Chemicals.

Due to the absence of indigenous manufacturing, the Indian HEMA market is driven by imports, meeting demand from local resin, paint, and polymer industries. Given the current market size and a healthy CAGR, India represents a potential opportunity for setting up a 10,000–40,000 TPA domestic production facility. Such a plant could substitute imports, reduce foreign exchange outflow, and meet the growing needs of coatings and specialty polymers sectors.

The price of HEMA varies significantly based on purity and grade. Technical-grade HEMA (used in coatings and adhesives) trades at around USD 2.5–4.0 per kg, whereas high-purity grades (for biomedical and optical applications) can exceed USD 10 per kg. Prices are sensitive to fluctuations in feedstock costs of methacrylic acid and ethylene oxide, as well as logistics and polymerization inhibitor requirements.

In conclusion, the Indian HEMA market is poised for steady growth in line with global trends. With increasing demand for adhesives, coatings, and advanced polymer materials, establishing domestic HEMA production presents a viable opportunity. This could position India as a regional supplier in South and Southeast Asia in the long term.

Table 2: Imports of HEMA

Quantity (MT)	***Total value (million USD)***	***Value per kg***	***Country***
444.54	70.75	160.00	Singapore
398.78	61.4	153.00	China
119.63	19.54	163.0	Japan
4.8	2.26	471.0	Switzerland
0.2	0.15	745.0	South Korea

1.5 Capacity of the HEMA manufacturing

The proposed capacity for the new 2-Hydroxyethyl Methacrylate (HEMA) plant is set at **20,000 tonnes per annum (TPA)**. At present, India imports approximately 7,000–8,000 tonnes of HEMA every year, as there are no indigenous producers. Since the estimated CAGR for HEMA is 6.5%, setting up a plant with capacity of 20,000 TPA, which is approximately half of the largest producer’s capacity, the plant can we economically feasible and it will be easy to sustain the market.

Chapter 2: Process selection

2.1 Literature survey

Upon doing a literature survey, it was found that two major industrial processes are used to produce HEMA economically. The first mention of the synthesis is dated to 1925. The two common methods of synthesis are (i) Reaction of Methacrylic acid with Ethylene Glycol and (ii) Reaction of Methacrylic acid with Ethylene Oxide (). Both reactions can lead to formation of Ethylene Glycol Di-methacrylate (abbreviated as **EDGMA).** Other reactions such as acyl chloride esterification method and dihydroxyl carbonic esterification reactions are also possible reactions to produce HEMA, however the conditions required for these reactions are achievable only at lab scale.

2.1.1 Epoxidation of Methacrylic Acid

Upon reacting ethylene oxide with methacrylic acid, the reaction producing HEMA proceeds by a nucleophilic substitution reaction at the (alpha) carbon. The reactivity of ethylene oxide plays a significant role in designing the process since it is a highly reactive and toxic compound. Hence the reaction is conducted at controlled temperatures and pressures. The common choices for a solvent media are any tertiary amines.

2.1.2. Trans-esterification of Ethylene Glycol

An alternative route involves the esterification of methacrylic acid with ethylene glycol. In this process, methacrylic acid reacts with an excess of ethylene glycol in the presence of an acid catalyst—commonly sulfuric acid, p-toluene sulfonic acid, or an acidic ion-exchange resin. The reaction proceeds via condensation, forming 2-hydroxyethyl methacrylate and water as a by-product. The formation of water is the major drawback of this method, since HEMA is completely miscible in water. Water also acts as a polymerization promotor and hence it leads to the formation of hydrogel.

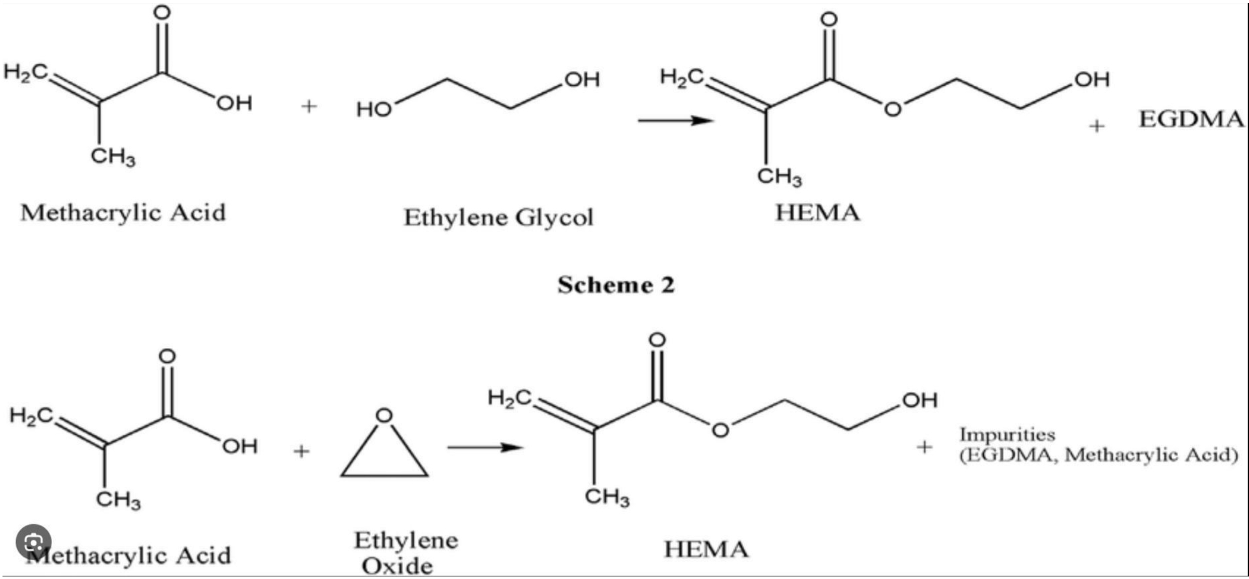


Figure 2: Synthesis routes for HEMA

2.2 Comparison of the two processes

Table 3: Comaprision of syntheses routes for HEMA

Process Variable	**Esterification Reaction**	**Epoxidation Reaction**
Pressure	Atmospheric Pressures	Vacuum of -0.99 – -0.75 Bar
Temperature	100-120 ℃	75 -90 ℃
Catalyst	P-methyl benzene sulfonic acid or ion exchange catalysts	Ferric Formate or Chromium Acetate
Reagent	Sulphuric acid	Tertiary amine such as pyridine, N,N-Dimethyl toluene etc.
Reaction time	1 – 3 hr	1.5 – 2 hr

Percentage yield	> 85%	> 99%
Polymerization inhibitor	Phenothiazine	MEHQ (hydroquinone monomethyl ether)
Impurities	Ethylene glycol dimethacrylate, Water	Ethylene glycol dimethacrylate

2.2.1 Analysis of the Esterification reaction (Changzhou Tianma group, 2012)

Advantages of the reaction are as follows-

1. The reaction is relatively simpler and it does not require using complex and hazardous raw materials, which also makes it safer than the ring opening reaction.
2. The reaction conditions are moderate.
3. There is no need to use expensive metal oxide catalysts.
4. Since ion-exchange resins can be used, the separation can be done much more easily.

Disadvantages of the reaction are as follows-

1. The reaction is limited by equilibrium.
2. The amount of side product (EGDMA) produced is much higher even if the amount of acid is controlled. There is a high change of a co-polymer formation if the amount of EGDMA is not controlled.
3. The water formed in this reaction is a major issue since HEMA is highly soluble in water. Water also promotes the polymerization reaction.
4. The yield and selectivity is also lower for this reaction

2.2.2 Analysis of Epoxidation of Methacrylic acid (Changzhou Hickory Chemistry Ltd. 2016)

Advantages of the reaction are as follows-

1. The reaction has better yield and selectivity and the amount of side-product is less.
2. There is no water formed as it is a direct one step synthesis.
3. The purification hence becomes simpler.

Disadvantages of the reaction are as follows-

1. Ethylene oxide is a highly flammable, highly carcinogenic and reactive material.
2. The reaction requires a very controlled environment. If the pressures or temperatures deviate even a little, there is a very high chance of a runaway reaction and a very major accident.

2.2.3 Comparison of the Routes based on economic potential

Economic potential is essentially the net difference between the price of the raw materials and the product. It can be used to compare the economic feasibility of two reactions to synthesize a chemical product.

The economic potential (EP) of a chemical product is given by the following formula

EP = Value of Products − Cost of Raw Materials and Utilities

or equivalently,

$$EP = \Sigma(m_p \times P_p) - [\Sigma(m_r \times P_r)]$$

Where-

m_p is the Mass (or molar) flow rate of product p

P_p is the Price of product p (Rs/kg or $/kg)

m_r is the Mass (or molar) flow rate of raw material r

P_r is the Price of raw material r

Using this, we can calculate the economic potential of the two routes. The route with more EP will be more feasible.

Table 4: Economic Potential of HEMA based on route using Ethylene Glycol

Chemical	**Moles**	**MW**	**Price/kg**	**price/mole**
EG	1	62	65	4.03
MAA	1	86	150	12.9
H2O	1	18	0	0
HEMA	1	130	160	20.8

For the esterification reaction,

$$EP = 20.8-12.9-4.03$$

$$EP = 3.87$$

For the esterification reaction,

$$EP = 20.8-12.9-4.03$$

$$EP = 3.87$$

Table 5: Economic Potential of HEMA based on route using Ethylene Oxide

Chemical	**Moles**	**MW**	**Price/kg**	**price/mole**
EO	1	44	86	3.784
MAA	1	86	150	12.9
HEMA	1	130	160	20.8

For the epoxidation reaction,

$$EP = 20.8-12.9-3.78$$

$$EP = 4.12$$

2.2.4 Final selection of the process

Based on the factors such as economic potential, raw materials used, yield, selectivity and the reaction conditions, the epoxidation route is more feasible for bulk production of HEMA. This route is also preferred globally.

2.3 Production of Methacrylic acid

Setting up a methacrylic acid (MAA) manufacturing plant alongside a 2-hydroxyethyl methacrylate (HEMA) plant is necessary for both economic and technical reasons. Economically, methacrylic acid is a key raw material for HEMA production through epoxidation; producing MAA onsite reduces raw material procurement costs, transportation expenses, and supply chain risks associated with market fluctuations or delays. This integration improves overall cost efficiency by enabling economies of scale in raw material handling, utilities, and infrastructure usage. Technically, the close coupling of methacrylic acid production with HEMA manufacturing streamlines process flow, enhances material purity control, and minimizes losses during intermediate transfer. Additionally, onsite MAA production provides flexibility in adjusting HEMA feed composition and scaling production volumes rapidly based on market demand. Together, these factors support a more robust, cost-effective, and technically optimized specialty monomer manufacturing facility essential for maintaining competitiveness in rapidly growing biomedical, coatings, and industrial segments where HEMA is widely employed.

A literature survey shows three major industrially relevant routes for producing methacrylic acid (MAA):

1. C4 oxidation (oxidation of C4 hydrocarbons such as isobutene/isobutane to methacrolein/methacrylic acid)
2. ACH (Acetone Cyanohydrin) process (conversion of acetone via acetone cyanohydrin to methacrylamide and then to MAA/MMA)
3. Oxidative dehydrogenation/dehydration of isobutyric acid (IBA) to MAA.

All three routes have been studied and commercialised to varying extents. The ACH route has been the historical commercial route for MMA/MAA, while C4 oxidation routes (via isobutene → methacrolein → MAA) and IBA oxidative dehydrogenation/dehydration have been investigated and implemented depending on feedstock availability and economics.

2.3.1 C4 Oxidation Route

The C4 oxidation route typically oxidises C4 streams (isobutene, isobutane, or mixtures from steam crackers) to methacrolein (MAL) and subsequently to methacrylic acid (MAA). It is a controlled catalytic oxidation of the C4 rich raw material. (Roma Chemical Ltd, 2006)

<u>Typical catalysts & conditions</u>

1. Catalysts: vanadium-phosphorus based catalysts (e.g., VPO, vanadium pyrophosphate), mixed metal oxides (Mo–V–P, V–Mo–Nb), and supported vanadium oxides.
2. Reaction steps: partial oxidation of isobutene/isobutane to methacrolein (gas-phase, oxidative catalysts), followed by oxidation of methacrolein to MAA.
3. Temperatures: commonly in the range ~300–450 °C for gas-phase oxidation catalysts for direct IBA oxidation; partial oxidation steps for C4 → methacrolein operate at lower temperatures (200–350 °C) depending on catalyst.
4. Pressure: near atmospheric to a few bar depending on reactor design.

<u>Yields & selectivity</u>

Industrial C4 oxidation processes can achieve high conversions with varying selectivity to MAA/MAL depending on catalyst and process optimisation. Reported selectivities to desired C4 oxidation products (MAL/MAA) are high but side oxidation to COx and other by-products occurs and must be minimized with appropriate catalyst and reactor design.

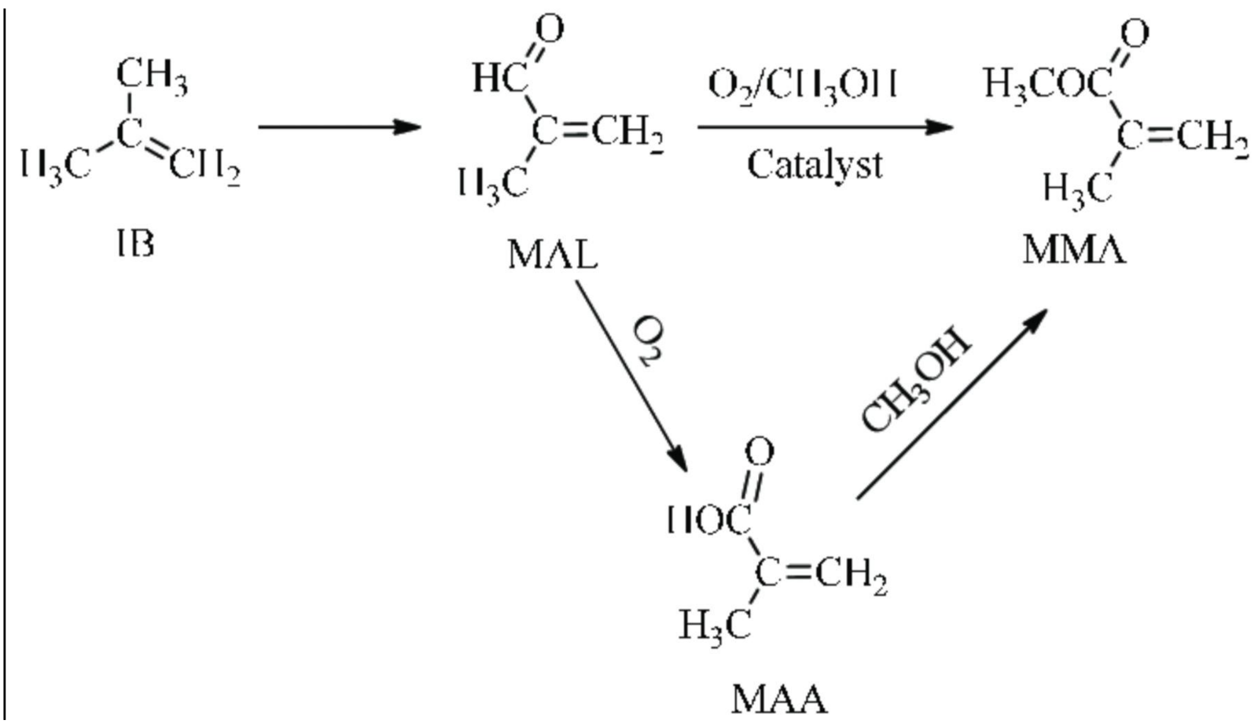


Figure 3: Partial oxidation of a C4 stream to give Methacrylic Acid

2.3.2 ACH (Acetone Cyanohydrin) Process

The ACH process converts acetone to acetone cyanohydrin (ACH) by reaction with hydrogen cyanide (HCN). ACH is then converted (hydrolysis followed by further steps) to methacrylamide or methacrylic derivatives and ultimately to MAA or MMA. Historically, the ACH route has been a major commercial route for methyl methacrylate (MMA) and related methacrylate chemicals. (Curtis, I. Carlson et. al., 2003)

Typical catalysts & conditions

1. ACH formation: acetone + HCN in the presence of a basic catalyst (phase transfer or base). Conditions are typically ambient to mild exothermic conditions; HCN handling and scrubbing systems are essential.
2. ACH hydrolysis / conversion: strongly acidic hydrolysis (traditionally sulfuric acid), or modern heterogeneous catalysts to avoid large quantities of waste acid. Temperatures and pressures vary with specific plant designs.

Yields & issues

The ACH route can give high yields to downstream methacrylic products when optimised. However, the process involves hydrogen cyanide—a highly toxic and hazardous reagent—and forms by-products such as acetone sulfonic acids, formamide (from HCN hydrolysis), and requires careful waste management.

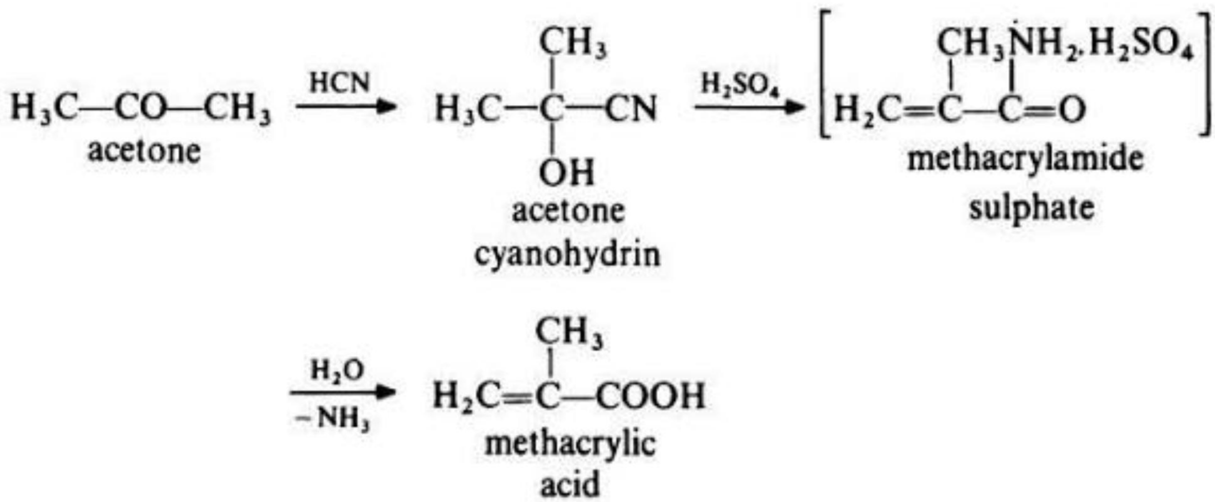


Figure 4: ACh process for Methacrylic acid production

2.3.3 Oxidative Dehydrogenation / Dehydration of Isobutyric Acid (IBA)

Isobutyric acid (IBA) can be oxidatively dehydrogenated/dehydrated to methacrylic acid in the gas phase over selective oxidation catalysts.

<u>Typical catalysts & conditions</u>

1. Catalysts: molybdenum- and vanadium-containing mixed oxides, heteropoly acids supported on silica, and modified molybdophosphoric materials.
2. Temperatures: typically 300–450 °C for gas-phase oxidative dehydrogenation.
3. Pressure: atmospheric to a few bar.

<u>Yields & selectivity</u>

Reported selectivities for IBA → MAA are moderate—literature and patents report selectivities in the range of roughly 60–75% under certain conditions. Side reactions and over-oxidation reduce overall selectivity and must be managed by catalyst choice and reaction conditions.

2.4 Comparison of the Three Processes

Table 6: Comparision of Syntheses routes for Methacrylic acid

Process Variable	**C4 Oxidation**	**ACH Process**	**IBA Dehydration /**

		(Acetone Cyanohydrin)	Oxidative Dehydrogenation
Pressure	Near atmospheric – 3 bar	1–2 bar (ACH formation), hydrolysis may use slightly higher pressure	Atmospheric – 2 bar
Temperature	200–450 °C (stepwise for oxidation)	0–50 °C for ACH formation; 60–120 °C for hydrolysis	300–450 °C
Catalyst	Vanadium-phosphorus oxides, Mo–V–P mixed oxides, supported V oxides	Acid catalysts (sulfuric acid, strong solid acids)	Molybdenum-vanadium mixed oxides, heteropoly acids
Reagent	Crude C4 stream (isobutene/isobutane), air/oxygen	Acetone + HCN (hazardous)	Isobutyric acid, O2 / air for oxidative route
Reaction time	Seconds to minutes (continuous gas-phase)	Minutes to hours depending on step	Minutes (gas-phase continuous)
Percentage yield	70–90% (to MAA, depending on selectivity)	80–95% (stoichiometric, high yield)	60–75% (moderate selectivity)

Impurities / By-products	CO, CO_2, maleic anhydride, polymeric tars, methacrolein residues	Acetone, ammonium salts, residual HCN, acetone sulfonic acid	COx, overoxidation products, polymeric residues

2.4.1 Analysis: C4 Oxidation Route

Advantages:

1. Compatible with crude C4 streams available at steam crackers (butenes, butadiene-rich cuts).
2. Can be configured as a continuous gas-phase oxidation process with high throughput.
3. Mature catalyst families (V–P, V–Mo, Mo–V–P) exist and have been optimised for selective partial oxidation steps.

Disadvantages and possible side reactions:

1. Over-oxidation to CO and CO2 (complete combustion) reduces selectivity.
2. Formation of by-products such as maleic anhydride, crotonaldehyde, acetaldehyde, and polymeric tars depending on feed composition and conditions.
3. Partial oxidation may produce methacrolein (MAL) as an intermediate—controlling the two-step oxidation is critical to avoid accumulation and side reactions.
4. Catalyst deactivation (coking, phase changes) and need for regeneration.
5. Presence of butadiene and other dienes in crude C4 can lead to undesired reactions (polymerisation, condensation) and require upstream purification.

2.4.2 Analysis: ACH Process

Advantages:

1. Historically proven route to MMA/MAA with high potential yields when optimised.

Disadvantages:

1. Requires hydrogen cyanide (HCN): highly toxic, poses acute safety hazards and complex environmental controls.
2. Side-products and acid wastes (if sulfuric acid used) can generate disposal and corrosion issues.
3. Modern improvements (heterogeneous catalysts for hydrolysis) reduce some waste but do not eliminate HCN hazards.

2.4.3 Analysis: Isobutyric Acid (IBA) Oxidation

Advantages:

1. Direct conversion from a C4 carboxylic acid; suitable where IBA feedstock is available.

Disadvantages:

1. Requires high temperatures (300–450 °C) and selective catalysts.
2. Selectivity to MAA is moderate (often 60–75%) with competing over-oxidation and by-product formation.

2.5 Feedstock Availability for Methacrylic Acid in India

Dahej is a major petrochemical hub with steam cracker and petrochemical complexes (OPaL/ONGC Petro additions and other operators) producing ethylene, propylene and C4 fractions as coproducts. Crude C4 streams (containing isobutene, butenes, butadiene, and butanes) are commonly produced at crackers and can be sourced from cracker operators, refineries, and C4 traders in the Dahej PCPIR/Dahej industrial cluster. Proximity to these crackers makes Dahej a favourable site for C4 oxidation processes.

2.6 Final Selection and Justification

Based on catalyst options, yields, continuous-operation compatibility, and feedstock availability in petrochemical clusters such as Dahej, the **C4 oxidation route** is selected for bulk production of methacrylic acid.

1. The C4 oxidation route integrates well with steam cracker crude C4 availability at Dahej, minimising feedstock logistics and cost.
2. It can be operated continuously with well-studied oxidation catalysts to achieve economically favourable yields and throughput.
3. The ACH route was rejected primarily because it requires hydrogen cyanide (HCN), which is highly hazardous and requires stringent safety measures and complex waste handling.
4. IBA oxidation was not selected due to moderately lower selectivity and the requirement for high-temperature operation and specialised catalyst systems.
5. The global competetors for methacrylic acid are also shifting to the C4 oxidation route due to the advantages of this process. Hence if we want to establish a plant for the long run, it is better to go for the C4 route.

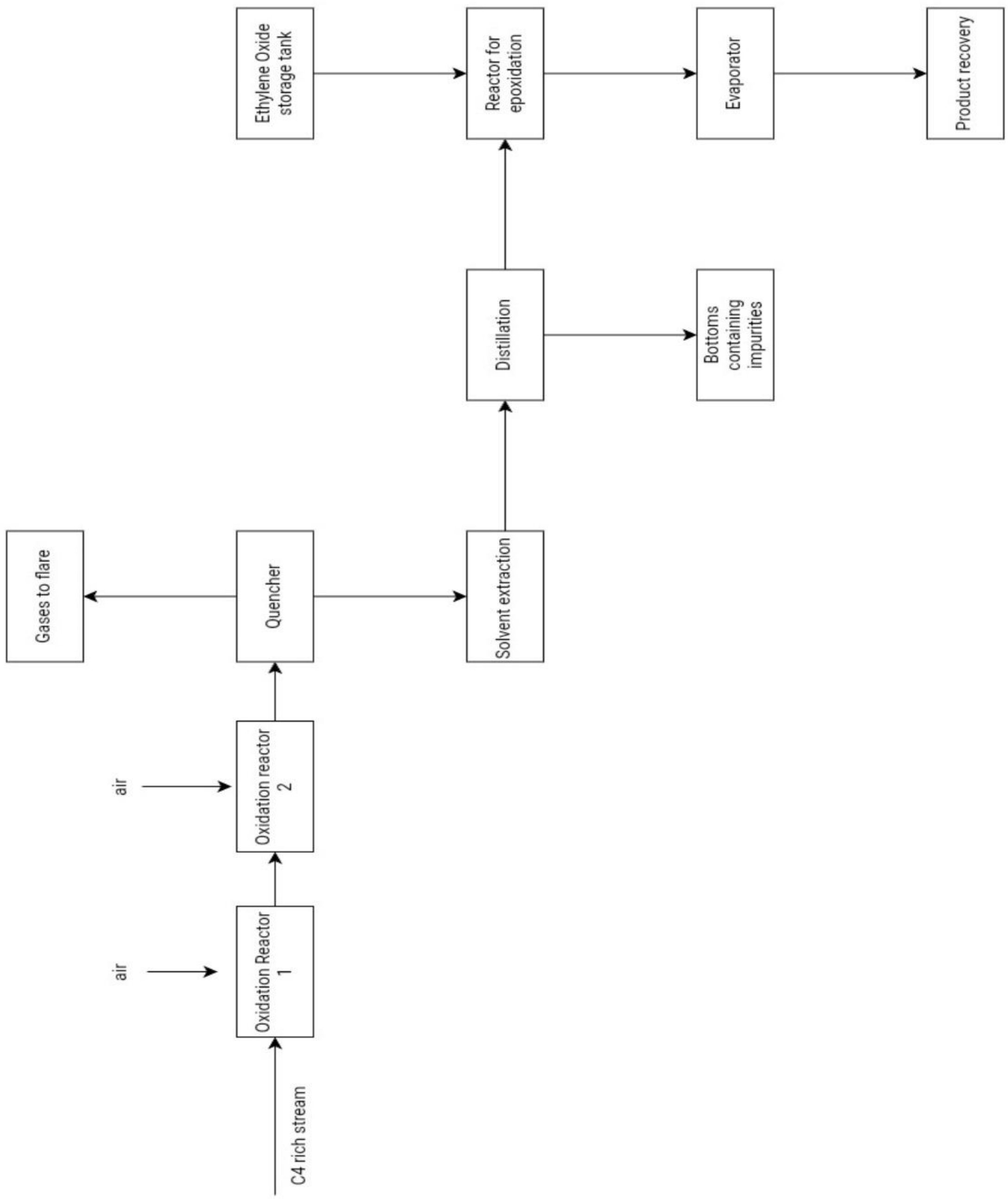


Figure 5: Block Flow Diagram for syntheses of HEMA and MAA

3: Site Selection

3.1: Factors for Site Selection

For a methacrylic acid (MAA) and 2-hydroxyethyl methacrylate (HEMA) plant, site selection is a critical strategic decision due to the hazardous, bulk nature of raw materials and the multi-step chemical processes involved. The core criteria are:

- Proximity to Petrochemical Feedstock: The catalytic oxidation process for MAA requires a reliable supply of C4 hydrocarbons and oxygen. Siting the plant near large petrochemical complexes minimizes feedstock transport costs and ensures operational continuity. For HEMA, access to methacrylic acid is essential for the epoxidation step; co-location within an established methacrylate cluster enables efficient intermediate handling.
- Logistics Infrastructure: MAA and HEMA plants demand robust inbound and outbound transport networks (road, rail, port) to handle bulk chemical supply and enable timely export of finished products. Integrated clusters with efficient connectivity help minimize delays and logistics expenses.
- Utilities and Steam Availability: These processes are energy-intensive, needing reliable electricity and process steam. Clusters like Dahej and Vapi have existing utility networks, and proximity to large chemical sites allows for possible steam purchase rather than captive generation.
- Water Supply and Effluent Management: Large quantities of water are required for reaction, cooling, and cleaning. Sites must have a sustainable water source and advanced effluent treatment facilities due to the hazardous nature of organic acids and solvents.
- Land and Regulatory Compliance: Sufficient land for plant, expansion, safety buffer zones, and environmental approvals. Gujarat and Maharashtra offer streamlined chemical sector land acquisition and clearances.
- Skilled Labor and Industrial Ecosystem: Availability of trained chemical engineers and technicians, plus proximity to suppliers of catalysts, solvents, and equipment.
- Ease of Doing Business: Favorable regulations, industrial cluster support services, and ease of obtaining hazardous chemical permits all impact setup and ongoing operations.

- Cluster Synergy: Co-location with allied industries (such as resins, adhesives, paints) in clusters facilitates feedstock availability, shared services, and downstream integration.

Justification for Exclusion of Certain Parameters

- Proximity to end-user market is less critical: MAA/HEMA are shelf-stable, and Indian demand is spread nationally; clusters are connected to export ports.
- Climate is not a major factor as chemicals degrade minimally under standard storage, and clusters have adequate warehousing.
- Effluent processing is ensured by cluster-scale treatment facilities; individual plants do not need to independently construct ETPs unless mandated.
- Housing and labor supply are sufficient in semi-urban chemical hubs, and relocation/hiring logistics are straightforward.

3.2: Site Comparison and Selection with Justification

Table 7: Site comparision

Parameter	Dahej PCPIR, Gujarat	Ankleshwar, Gujarat	Vapi, Gujarat	Thane, Maharashtra
Petrochemical Complex	GNFC, Reliance, ONGC	GNFC, Narmada Chemicals	Diverse Chemicals, SME units	Deepak Fertilizers, legacy clusters
Transport Links	Major port, rail, road	Rail, road, moderate port	Rail, road, Mumbai port	Rail, road, Mumbai port
Utility Cost	Low, cluster tariffs	Low, cluster tariffs	Moderate, SME rates	Higher tariffs
Steam Availability	Cluster steam supply	Cluster steam, captive	Cluster steam, captive	Captive only

Water Cost (Rs/kL)	15–18	18–20	19–22	24–28
Land Cost (Lakh/acre)	16–22	13–20	15–24	38–45
Labor Cost (Rs/day)	480	490	520	560

Table 8: Weightage of each parameter

Parameter	**Weight (0–10)**
Petrochemical complex	10
Transport links	9
Utility/steam	8
Water cost	7
Land cost	8
Labor cost	5
Ease of doing business	8
Cluster synergy	8

Table 9: The grading of Sites Based on Parameters

Parameter	**Dahej**	**Ankleshwar**	**Vapi**	**Thane**

Petrochemical complex	10	9	8	7
Transport links	10	8	8	8
Utility/steam	9	8	8	6
Water cost	9	8	8	6
Land cost	8	9	7	5
Labor cost	7	7	7	8
Ease of doing business	9	9	7	7
Cluster synergy	10	9	9	8

Table 10: Weighted Average of Parameters

Weighted Parameter	Dahej	Ankleshwar	Vapi	Thane
Petrochemical complex	100	90	80	70
Transport links	90	72	72	72
Utility/steam	72	64	64	48
Water cost	63	56	56	42
Land cost	64	72	56	40
Labor cost	35	35	35	40

Ease of doing business	72	72	56	56
Cluster synergy	80	72	72	64
Total weighted score (out of 10)	6.9	6.6	5.9	5.0

Based on the weighted score comparison, **Dahej PCPIR** in Gujarat emerges as the optimal location for setting up a methacrylic acid and 2-hydroxyethyl methacrylate plant. This site offers direct access to petrochemical feedstocks, export ports, cluster steam/utilities, cost-effective land, and a mature chemical manufacturing ecosystem, making it ideal for specialty monomer production and expansion.

4: Thermodynamic Feasibility

The thermodynamic feasibility of any process is characterized by the Gibbs free energy of that process at the given condition. The Gibbs free energy or ΔG for any process should be negative for the process to be feasible. The standard Gibbs free energy or Δ°G is the ΔG at 25 °C or 298 K. The Table below lists the standard Gibbs energy of formation of the main species involved in the synthesis of HEMA using the chosen process. (All the thermodynamic values have been taken from Yaws C.L. ,"Physical Properties Handbook", McGraw Hill publications)

Table 11: Standard Gibbs energy of formation

Species	**Δ°G formation (kJ/mol)**
Isobutylene	58.07
Methacrolein	-57.6
Methacrylic Acid	-287.72 for gas phase, -203.65 for liquid phase
Water	-13.1
Ethylene Oxide	-291.81
2-Hydroxyethyl Methacrylate	-228.64

The reactions occurring for forming HEMA are as follows-

$C_4H_8(g)(Isobutene)+O2(g) \rightarrow C_3H_6O(g)(Methacrolein)+H_2O(g)$ Reaction 1 1;

$C_4H_6O+\frac{1}{2}O_2 \rightarrow C_4H_6O_2(g)(Methacrylic\ Acid)$Reaction 2;

$C_4H_6O_2(l)+C_2H_4O(g)(Ethylene\,Oxide) \rightarrow C_6H_{10}O_3(l)(HEMA)$Reaction 3;

The $\Delta G°$ of any reaction is given by the formula

$\Delta G^0 = \Sigma\, v_i G_{fi}$

where n is the stochimetric coefficient of species “i” and Gf is the Gibbs energy of formation for the species.

Using this relation, we obtain the Gibbs energy of formation at standard conditions for all the three major reactions. The results are in the table given below-

Table 12: Gibbs energy of formation for the reactions

Reaction no.	Gibbs energy (kJ/mol)
1	-344.31
2	-230.12
3	-75.06

Since for all three reactions the ΔG° is negative, we conclude that all the reactions are thermodynamically feasible. Since the ΔG° is not zero, the reactions are not limited by equilibrium.

5: Kinetics of the Reaction

5.1 Partial oxidation of Isobutene and Methacrolein

The partial oxidation of both Isobutene and Methacrolein follow the Mars-van Krevelen mechanism. The Mars-van Krevelen mechanism is when the hydrocarbon gas molecule reacts with the adsorbed oxygen molecule in the reactant pore. The mechanism is as follows-

$$A(g) + X \rightarrow AX$$

$$B(g) + AX \rightarrow CX$$

$$CX \rightarrow C(g) + X$$

where, A represents Oxygen, X represents the vacant site on the catalyst, B represents the hydrocarbon species and C is the product molecule. The reaction has a characteristic factor of φ which the desorption rate of Oxygen from the catalyst site.

5.1.1 Kinetic parameters for partial oxidation of Isobutene to Methacrolein

Molybednum based catalysts provide the highest selectivity for this reaction. Based on the work done by Hoornaerts et. Al in their 1997 paper titled “Modification of kinetic parameters by action of oxygen spillover in selective oxidation of isobutene to methacrolein” the rate equation for this reaction is given as

$$r_{MAC} = \left(\frac{k_{red}\ P_{C4}^{x}\ k_{ox}\ P_{O2}^{y}}{k_{red}\ P_{C4}^{x} + k_{ox}\ P_{O2}^{y}} \right)$$

The values of the kinetic parameters are given in Table 13.

Table 13: Kinetic parameters for reaction 1, Benyahia et. al

Temperature (K)	k_red (mol atm^{-} s^{-} m^{-2})	k_ox (mol atm^{-} s^{-} m^{-2})	X	Y	ψ
623	8.88e-11	0.92e-11	0.03	0.7	0.16
643	16.82e-11	1.72e-11	0.25	0.77	0.09
663	26.6e-11	3.47e-11	0.33	0.7	0.12

The activation energy for catalyst reduction is 52 kJ/mol and for reoxidation is 142 kJ/mol. The constant K0 for the kinetic parameter is chosen to the parameters at 643 K due to the low psi value and Arhenius equation is used to calculate the values at any temperature.

5.1.2 Kinetics parameters for the partial oxidation of Methacrolein

Heteropoly acid catalysts are used for facilitating these reactions. They are usually phospho-Molybedate complexes. The rate equation for the oxidation of Methacrolein is given by

$$-r = \frac{K_{O_2} P_{O_2}^{n} \left(1 + P_{H_2O}^{m}\right)^2}{K_{O_2} P_{O_2}^{n} \left(1 + P_{H_2O}^{m}\right) + 0.5\, K P_{MAC}}$$

where, n = 0.7, m =0.65 and the values of the kinetic parameters are described in the following table

Table 14: Kinetic parameters for reaction 2

Kinetic Parameter	Value (mol/hkgcatPa^x)	Activation energy (kJ/mol)
Ko2	5.35E-06	99.99
K	7.6E-04	154.4

5.2 Esterification of Methacrylic acid with Ethylene Oxide

The rate equation for this reaction is given by

$$-r_{EO}=\frac{K_1 K_2 k_3 [EO][MA]}{1+K_1[EO]+K_1 K_2[EO][MA]+K_4^{-1}[HEMA]}$$

where, [Cx] is the concentration of the species *"x"* and the value of the kinetic parameters are given in the table below.

Table 15: Kinetic parameters for reaction 3

Kinetic Parameter	Value at 50 °C	Value at 60 °C	Value at 70 °C	Value at 80 °C	Value at 90 °C	Activation energy (kJ/mol)

K1	6.654	3.909	2.368	1.478	0.944	33.53
K2	0.396	0.368	0.344	0.322	0.304	47.62
k3	0.0887	0.183	0.459	1.095	2.488	54.11
K4	0.08	0.19	0.428	0.921	1.901	77.21

The catalysts for this reaction include iron or chromium based catalysts. The use of a tertiary amine also helps in promoting this reaction.

6: Material Balance

Material balance is essentially applying the law of conservation of mass to the reactor system. We can do an overall balance for the reactor by using the known conversion values. The conversion of Isobutene in reactor 1 is 99.6% and the yield of Methacrolein is 79.6%. Similarly for reactor 2, the conversion of methacrolein is 78% and the selectivity is 85%. Low conversions of methacrolein are preferred since at conversions higher than 78%, the selectivity to Methacrylic acid decreases. For the esterification, the conversion with respect to methacrylic acid is 99.5%. Here, ethylene oxide is the limiting reactant due to its high reactivity.

For given production rate of 8,000 tonnes per annum, we have the production rate of HEMA to be 1 tonnes per hour or 19.23 kmol/h. Using this we back calculate the amount of Isobutene required. The flow rate of methacrylic acid production will be approximately be the same. Using patent no. CN114057571A as a reference, we get the value of isobutylene molar flow rate as 41.75 kmol/h. The calculations done for this is as follows -

$$F_{MAA} = F_{MAL} \text{ x yield of MAA} = F_{MAL} \text{ x } 0.663$$

$$F_{MAL} = 33.23 \text{ kmol/h}$$

$$F_{MAL} = F_{IB} \text{ x yield of MAL} = F_{IB} \text{ X } 0.796$$

$$F_{IB} = 41.75 \text{ kmol/h.}$$

Where, F_x is the molar flow rate of the species "x",

MAL is abbreviation of Methacrolein

MAA is abbreviation of Methacrylic acid.

Now from literature, it is seen that the ideal range for the molar ratio of IB to Oxygen to steam is

1: 2: 1.5. Using this composition, we can easily calculate the inlet flow rate to reactor R201 and its composition. This stream is labelled as stream 4 in the Process flow diagram.

Table 16: Feed to the reactor

	mol%	**kmol/h**	**mw**	**kg/h**
IB	8.63	41.75	56.00	2,338.00

O2	17.25	83.50	32.00	2,672.00
N2	61.18	296.05	28.00	8,289.27
steam	12.94	62.63	18.00	1,127.25
Total	0.00	483.92		14,426.52

The mass balance of the outlet stream from the reactor and its succeeding units has been carried out and found out based on the compositions mentioned in the patent CN114057571A. For the species whose molar composition in a stream was unknown, a basic elemental balance was used to find out the same. For each stream it was ensured that the mass in was equal to the mass out. Given below is the list of the all the units in the plant.

Table 17: List of units in the plant

Equipment	**Description**	**Type**
R201	Oxidation reactor	Multi tubular packer bed
R202	Oxidation reactor	Multi tubular packer bed
C301	Condenser	

C203	Crystallizer	
Filter		
T204	Extraction column	Packed column
T302	Distillation column	Plate column
T303	Distillation column	Plate column
T304	Distillation column	packed
T204	Distillation column	packed
R305	Esterification reactor	Cstr (gas-liq)
R306	Esterification reactor	Cstr (gas-liq)
R307	Esterification reactor	Cstr (gas-liq)
R308	Esterification reactor	Cstr (gas-liq)
T306	Distillation column	packed

6.1 Reactor system Material balance

For the first oxidation reactor R201, the feed is Isobutene, Steam and air mixture which is

intially heated to 350 °C and sent to the reactor. The reactor is a multi-tubular packed bed reactor which is said to operate isothermally. The coolant used is a thermic salt. The conversion of Isobutene is 99.4% and the selectivity to methacrolein is 80%. Using these values, we can calculate the amount of Methacrolein that will be formed. Let molar flow rate of isobutene be denoted by FIB0 and the molar flow rate of methacrolein be FMAC.

So,

$$FMAC = 0.994 \times 0.8 \times FIB$$

$$FMAC = 0.994 \times 0.8 \times 41.75$$

$$FMAC = 33.23 \ kmol/h$$

Patent CN114057571A mentions the composition of the reactor outlet but the value of steam and oxygen was not given. So using the elemental balance for carbon and oxygen, the outlet flow rate of steam and water was also calculated. Nitrogen is an inert gas so the moles of nitrogen in and moles out will be the same. Table 18 gives the outlet of the reactor 1.

Table 18: Stream Composition of outlet of reactor 201

	Mol%	**Molar flow rate (kmol/h)**	**MW**	**Mass flow rate (kg/h)**	**wt%**
IB	0.03	0.17	56.00	3.61	0
MAC	7.05	34.47	70.00	2,412.66	16
CO	1.05	5.12	28.00	143.46	0

MAA	0.30	1.45	86.00	125.04	0
Acetic acid	0.30	1.45	60.00	87.24	0
acetone	0.17	0.83	58.00	48.19	0
acetaldehyde	0.30	1.45	44.00	63.97	0
acrolein	0.06	0.28	56.00	15.51	0
formaldehyde	0.04	0.21	30.00	6.23	0
acrylic acid	0.04	0.21	72.00	14.95	0
O2	5.39	26.35	32.00	843.20	5
N2	60.55	296.05	28.00	8,289.27	57
CO2	1.55	7.56	44.00	332.50	2
steam	23.19	113.37	18.00	2,040.70	14
formic acid	0.00	0.00	46.00	0.00	0

total	100.00	488.97		14,426.52	100

Similarly, for the second oxidation reactor R202, the conversion of methacrolein is 78 % and the selectivity to methacrylic acid is 85 %. So the molar flow rate of methacrylic acid FMAA will be given by -

$$FMAA = 0.85 \times 0.78 \times FMAC$$

$$FMAA = 0.85 \times 0.78 \times 33.23$$

$$FMAA = 23.81\ kmol/h$$

Again using the compositions given in the patent, the outlet of the reactor R202 was also calculated. The makeup air that was to be added to the reactor, was calculated by elemental balance of Nitrogen, Oxygen and Carbon. The stream composition of the reactor outlet is given below. The mass flow rate of the makeup air is 19,690.91 kg/h.

Table 19: Stream composition of outlet of Reactor R202

	Mol%	**Molar flow rate (kmol/h)**	**MW**	**Mass flow rate (kg/h)**	**wt%**
MAC	0.70	7.58	70.00	530.78	1.56
CO	1.00	10.83	28.00	303.31	0.89
MAA	2.28	24.70	86.00	2,124.00	6.23

Acetic ac	0.31	3.36	60.00	201.48	0.59
acetone	0.12	1.30	58.00	75.39	0.22
acetaldehyde	0.21	2.27	44.00	100.09	0.29
acrolein	0.04	0.43	56.00	24.26	0.07
formaldehyde	0.03	0.32	30.00	9.75	0.03
acrylic acid	0.03	0.32	72.00	23.40	0.07
oxygen	6.50	70.41	32.00	2,253.12	6.60
steam	18.98	205.64	18.00	3,701.49	10.85
CO2	3.32	36.00	44.00	1,584.10	4.64
N2	76.43	827.86	28.00	23,180.18	67.94
TPA	0.003	0.037	166.00	6.07	0.02
Total		1,083.23		34,117.43	

6.2 Condenser

Before sending the outlet gas from the reactor to the condesner, it is cooled to 230 °C and then sent to the quencher. The condensation occurs at 80 °C, this separates methacrolein, acrolein and lighter carbon compounds in the gas phase while the liquid phase will contain some amount of Methacrloein dissolved. The liquid phase will also contain Methacrylic acid, Methacrolein, Impurity which is Terepthallic acid, acrylic acid, Acetic acid and large quantity of water. For all the separation steps after the reactor, 1000 ppm of a polymerisation inhibitor (in our case MEHQ) needs to sprayed throughout the columns to avoid the formation of polymers. The liquid stream composition is given below. The gas stream is sent to the flare for combustion.

Table 20: Liquid stream composition

	Mol%	Molar flow rate(kmol/h)	MW	Mass flow rate (kg/h)	wt%
MAA	10.54	24.70	86.00	2,124.00	35.00
MAC	0.07	0.17	70.00	12.14	0.20
Impurity	0.02	0.04	166.00	6.07	0.10
acrylic acid	0.14	0.32	72.00	23.40	0.39
acetic acid	1.43	3.36	60.00	201.48	3.32
water	87.79	205.64	18.00	3,701.49	60.99
Total		234.23		6,068.58	100.00

The liquid stream is then sent to the crystallizer to crystal out TPA at 13 °C. The entire TPA is crystalled out and the liquid stream outlet of the crystallizer is sent to an extraction column.

6.3 Extraction column

Since the compounds have the ability to polymerize, there is no data available for the Liquid-liquid equilibrium data. So for the extraction column, the interaction parameters were calculated using UNIFAC and the model used for LLE was NRTL. Solvent chosen for extraction was n-Butyl Acetate (NBA) which has the tendency to go nto the organic phase and has low solubility in water. The python code to calculate the unifac parameters is attached in Appendix 1.

The amount of solvent requires was 20000 kg/h. The composition of both phases is given below

Table 21: LLE outlet streams

Component	**Feed (Aqueous)**	**Solvent In (n-BuAc)**	**Extract (Organic out)**	**Raffin (Aqueou**
MAA (Methacrylic acid)	**2,124.0**	0.0	**2,066.7**	**57.3**
Methacrolein (MAC)	12.0	0.0	12.0	**0.0**
Acrylic acid	23.0	0.0	22.9	**0.1**
Acetic acid	201.0	0.0	201.0	**0.0**
Water	3,701.0	0.0	986.2	2,714

n-Butyl Acetate (solvent)	0.0	20,001.3	18,568.0	1,433
Total mass flow (kg/h)	**6,061.0**	**20,001.3**	**21,856.8**	**4,205**

The stream tables are attached in annexure 5 for rest of the equipments.

7: Energy balance

The values of specific heat and the enthalpy of formation were taken from Yaws' Handbook of physical Properties. Both were functions of temperature. Based on these values, energy balance was done for all the three reactors. The enthalpy of formation (Hf) is given by

$$Hf = A + BT + CT^2,$$

where Hf is in kJ/mol, T is in K and the component is in gas phase.

Table 22: Regression coefficients for enthapy of formation

Component	A	B	C
CO	-112.19	8.12E-03	-8.04E-06
CO2	-393.422	1.50E-04	-1.39E-06
Formaldehyde	-109.671	-2.32E-02	7.87E-06
acetic acid	-422.584	-4.84E+00	2.34E-05
Acetaldehyde	-154.122	-4.72E-02	2.03E-05
Acrolein	-66.702	-5.95E-02	3.87E-05
Acryclic acid	-325.038	-4.41E-02	2.09E-05
Acetone	-199.175	-7.15E-02	3.25E-05

Methacrolein	-90.882	-8.68E-02	5.28E-05
Methacrylic acid	-430.04	-4.75E-02	3.47E-04
Impurity (TPA)	-688.693	-1.13E-01	4.93E-05
Water	-238.41	-1.23E-02	2.77E-06
Ethylene Oxide	-38.88	-5.40E-02	2.56E-05
Isobutylene	4.126	-8.16E-02	3.66E-05

Chapter 8: Equipment Sizing

For the equipments given in Table 18, the complete mechanical design was done for Reactor R201 and the extraction column T204.

8.1 Reactor R201

The feed to the reactor is stream 4, containing Isobutene (IB), steam and air entering at 350 °C. The reactor is a multi-tubular packed bed reactor, where the catalyst used is a MoVBiCsRh oxide. The outlet contains unreacted Isobutene, Methacrolein (MAC), methacrylic acid (MAA), acetone, formaldehyde, acetaldehyde, acetic acid, acrylic acid, acrolein, carbon dioxide, carbon monoxide, nitrogen, oxygen and steam. Based on this, the following reactions can be said to have occurred in the reactor;

$$IB + O_2 \rightarrow MAC + H_2O \;\ldots..\; (1)$$

$$MAC + 0.5\,O_2 \rightarrow MAA \;\ldots..\; (2)$$

$$MAC + 4O_2 \rightarrow \text{Acetic acid} + CO_2 + 2H_2O \;\ldots..\; (3)$$

$$MAC + 3O_2 \rightarrow \text{Acryclic acid} + H_2O \;\ldots..\; (4)$$

$$MAC + 2.5O_2 \rightarrow \text{Acrolein} + CO_2 + H_2O \;\ldots..\; (5)$$

$$MAC + 3O_2 \rightarrow \text{Acetaldehyde} + 2CO + H_2O \;\ldots..\; (6)$$

$$IB + 6O_2 \rightarrow 4CO_2 + 4\,H_2O \;\ldots..\; (7)$$

$$IB + 4O_2 \rightarrow 4CO + 4\,H_2O \;\ldots..\; (8)$$

$$MAC + 1.5\,O_2 \rightarrow \text{Acroelin} + CO_2 + H_2O \;\ldots..\; (9)$$

$$MAC + 5O_2 \rightarrow 4CO_2 + 3H_2O \;\ldots..\; (10)$$

$$MAC + 3O_2 \rightarrow 4CO + 3H_2O \;\ldots..\; (11)$$

$$IB + O_2 \rightarrow \text{Acetone} + \text{Formaldehyde} \;\ldots..\; (12)$$

$$\text{Formaldehyde} + 0.5O_2 \rightarrow CO \;\ldots..\; (13)$$

$$\text{Formaldehyde} + O_2 \rightarrow CO_2 + H_2O \;\ldots..\; (14)$$

8.1.1 Extent of Reactions

Based on the known outlet and inlet, to determine the actual reactions occurring in the reactor, for all the reactions, the extent of reactions was determined where extent of reactions ζ is given by -

where, v_{ij} is the stochiometric coefficient of species "i" in reaction "j". For each of the 14 reactions, the extent was calculated using the python code mentioned in Annexure 1.

From the computational calculations, the following extent of reactions were obtained

Table 23: Extent of the reactions occuring in R201

Sr. no	**Reaction**	**Extent of the reaction**
1	IB + O2 → MAC + H2O	38.7000
2	IB + 1.5 O2 → MAA + H2O	1.5191
3	IB +5O2 → Acetic acid + CO2 + 3H2O	1.1094
4	IB + 3O2 → Acrylic acid + 2H2O	0.0303
5	IB + 2.5O2 → Acrolein + CO2 +2 H2O	0.0000
6	IB + 3O2 → Acetaldehyde + 2CO + 2H2O	1.1094
7	IB + 6O2 → 4CO2 + 4 H2O	1.1650
8	IB + 4O2 → 4CO + 4 H2O	0.6542
9	IB + 2.5 O2 → Acroelin + CO2 + 2H2O	0.5194
10	MAC + 5O2 → 4CO2 + 3H2O	0.0000
11	MAC + 3O2 → 4CO + 3H2O	0.0000
12	IB + O2 → Acetone + Formaldehyde	0.6115

Sr. no	Reaction	Extent of the reaction
13	Formaldehyde + 0.5O2 → co	0.5718
14	Formaldehyde + O2 → CO2 + H2O	0.0000

So we can ignore reaction numbers 5, 10, 11 and 14 for the reactor design part.

8.1.2 Mass balance across the reactor

To have a accurate mass balance, it is necessary to obtain the flow rates of the products in terms of the inlet flow rates, selectivity and the conversion. So if the inlet flow rate of the limiting reactant (for our case Isobutene) is given by F_{IB0}, for oxygen it is given by F_{O20}, for nitrogen it is given by F_{inerts} and for steam it is given by F_{H2O0}, the selectivities for methacrolein, side products, carbon monoxide and carbon dioxide are Ya, Yb, Yc and Yd respectively and the conversion of isobutene is given by Xa, the mass balance can be obtained in the table 28. Here Ya is 0.82, Yb is 0.1, Yc is 0.03 and Yd is 0.05. Since the reactor is continuous and the rate equations will use the partial pressures and concentrations, this is a necessary step to do.

Table 24: Mass balance across the reactor

Species	initial	At any point in the pbr
MAC	0	FIB0*Xa*Ya
Side products	0	FIB0*Xa*Yb

CO	0	FIB0*Xa*Yc*4
CO2	0	FIB0*Xa*Yd*4
IB	FIB0	FIB0*(1-Xa)
O2	FO20	FO20 – (FIB0*Xa*Ya – FIB0*4.75*xa*Yb – FIB0*Xa*Yc - FIB0*Xa*Yd*6/4)
INERTS	Finerts	Finerts
H2O	FH2O0	FH2O0 + FIB0*Xa*Ya + Fib0*Xa*Yb + FiB0*Xa*Yc*4 + FiB0*Xa*Yd*4
Total	FT0	FT0 + FIB0Xa(6Yc + 5.5Yd – 3.75Yb)

8.1.3 Process design

For the reactions occurring in reactor, most are consuming isobutene as the primary step. The side products are formed by a series of intermediate reactions, but the rate limiting step for all the reactions is the formation of a C4 carbo-cation (Bauge et. Al., 1999). The summary of the kinetics for the reactions is given in table Table 25: Summary of reaction kinetics.

Table 25: Summary of reaction kinetics

Reaction	Rate equation	Parameter values
IB + O2 → MAC + H2O	$-r_{ib}=s'\frac{k_{ox}P_{o2}^{y}k_{red}P_{ib}^{x}}{k_{ox}P_{o2}^{y}+k_{red}P_{ib}^{x}}$	K_{ox} = 7.23e^{-12} exp(142e^{03}/RT) K_{red} = 3.22e^{-11} exp(52e^{03}/RT) x = 0.8 y = 0.37 s' = 3.2e^{03} m2/kg P is the partial pressure in atm
IB + 4O2 → 4CO + 4H2O	$-r_{ib}=kC_{ib}C_{o2}^{0.5}$	K = 2.3e^{-03}exp(-158.1e^{03}/RT)
IB + 6O2 → 4CO2 + 4 H2O	$-r_{ib}=kC_{ib}C_{o2}^{0.5}$	K = 2.3e^{-03}exp(-158.1e^{03}/RT)
IB + 4.75O2 → Side products + H2O	$-r_{ib}=kC_{ib}C_{o2}^{0.5}$	K = 1.7e$^{-2.5}$ T$^{1.83}$exp(-309.6e^{03}/RT)

The design equation for the reactor will be given by

$$\frac{dW}{dXa}=\frac{FA0}{-R_{ib}}$$

Here $-R_{ib}$ is the total consumption of isobutene for all the reactions were it is a reactant.

Solving this ode, the weight of the catalyst is obtained as 21618.89 kg. (Code attached in annexure 2)

8.1.4 Catalyst properties

The particle density of the catalyst is said to be 2400 kg/m^3 and the voidage is 21%. The specific area of the catalyst is 3.4 m2/g and the diameter is 30 mm (Patents WO2012141076A1 and DE-OS 27 35 414)

8.1.5 Mechanical design

Using the weight of the catalyst and the bulk density, we obtain the volume of the bed as 11.38 m3. Assuming a fixed height to diameter or **H/D** ratio for the catalyst bed, we can find the Height of the bed that will be required. Assuming H/D to be 8 (patents say the H/D ratio has the range of 3-20, US20070049769A1), we get the Height of the bed as 9.75 m. Assuming the support space as 15%, the height of the cylinder can be found:

Hbed $\times$ (1 + 0.15) = 9.75 (1.15) = 11.21 m.

For the safe design, the design pressure and temperature should be 1.1 and 1.5 time the operating pressure and temperature. Hence the design pressure is 2.42 bar and design temperature will be 934.73 K.

The volumetric flow rate of the inlet is calculated by using ideal gas law and it comes out to be 3.17 m3/s. Hence the space time for the reactor is the bed volume divided by the volumetric feed flow rate which comes to be 4.13 sec.

Table 26: Reactor design parameters

Reactor Sizing Parameters					
V_{bed} (m^3)	$Diameter\,(m)$	$H_{bed}(m)$	$H_{cylinder}(m)$	$A_{cross}(m^2)$	$V_{cylinder}$ (m^3)

11.38	1.22	9.75	11.21	1.17	13.09

8.1.5.1 Materials of construction

When selecting a material of construction for a multitubular packed bed reactor performing the partial oxidation of isobutene to methacrolein at 2.2 bar absolute and 350 °C, SS316L (austenitic stainless steel, low carbon) presents a highly suitable choice. The process environment—combining an oxidizing organic feed, elevated temperature, and the presence of steam or water byproduct—demands robust resistance to pitting, crevice corrosion, and general oxidation, particularly given that bismuth molybdate catalysts often operate with slight halide or sulfur impurities that can accelerate localized attack. Unlike SS304, SS316L contains 2–3% molybdenum, which significantly enhances its resistance to chloride-induced stress corrosion cracking and pitting in moist oxidative atmospheres. Its low carbon content (<0.03%) prevents carbide precipitation at grain boundaries during welding, ensuring stable corrosion resistance without sensitization—a critical advantage for fabricating complex multitubular bundles with tube-to-tubesheet welds. Mechanically, SS316L retains excellent creep resistance and tensile strength at 350 °C, while its austenitic structure provides the thermal expansion compatibility needed between tubes and shell during cyclic start-ups. Although more costly than carbon steel or SS304, the superior long-term durability against uniform and localized corrosion under these specific conditions justifies SS316L as a reliable, maintainable, and safe metallurgical choice for this service.

Table 27: Properties of SS316L

Property	**Value**
Common standard	BS1501 Part 4 Grade 316L

AISI type	316L
Composition	0.08% C, 2% Mn, 8 − 11% Ni, 17.5 − 20% Cr,≤0.03% C, ≤2.0% Mn, 10–14% Ni, 16–18% Cr, 2–3% Mo
Thermal Conductivity (W/m°C)	18.4
Melting Point(°C)	1370-1400
Density (kg/m^3)	7980
Tensile Strength at 380°C (MPa)	114

8.1.5.2 Tube design

Assuming a tube outer diameter of 31.75, length 12 m and a trial thickness of 2mm, we can calculate the thickness using the thickness for a vessel under pressure. We calculate the ratio of outer diameter to assumed thickness and the ratio of length to the outer diameter to get the value of factor A and factor B to get the allowable pressure value. If allowable pressure is more than design pressure, the thickness will be adequate. The following results are obtained from the calculations -

Table 28: Tube design parameters

Design parameter	**Value**	**Units**
Trial thickness of tube	2	mm

Do/t	15.88	NA
L/Do	353.19	NA
Factor A	5.00E-05	NA
Factor B	5.50	mPa
Allowable P	0.46	mPa

Hence the allowable pressure is 4.6 bar which is more than our design pressure.

Now to get the number of tubes, we calculate the inner volume of the tube using the formula V = $\pi r_i^2 L$ which comes to be 7.26 mm3. The volume of the catalyst should be equal to the number of tubes into the volume of one tube. Hence we get the number of tubes as 1242.

The cross section area of one tube will be calculated as 604.81 mm2. Now using this we can get the velocity in one tube using the formula

Q = Area into velocity into number of tubes

So velocity of gas in one tube will be 4.22 m/s.

Using the formula for baffle diameter for square pitch, we get the baffle diameter as 1.81m, if we assume baffle diameter to be 95% of shell diameter, the shell diameter is 1.91m.

8.1.5.3 Shell design

The shell side will have the heat transfer fluid. The chosen thermic fluid in this case is LiNaKNO3 salt. It will be passed in the molten form. The melting point is 451 K. The calculations for the fluid on the shell side are given below in Table 29: Heat transfer fluid properties

Table 29: Heat transfer fluid properties

Property	Value	Units
Specific heat	1.65	kJ/kgK
Inlet temperature	401	K
Maximum allowed outlet Temperature	653.37	K
Velocity	0.80	m/s
Density	2700	kg/m3
Flow rate	7.80E-02	m3/s
Required heat transfer rate	6.90	MW
Calculated Temperature difference	19.85	K

8.1.6 Heat transfer calculations

Using Dittus Boelter equation we get the heat trasnfer coeffiecient on the shell side as 211.08 W/m2K and on the tube side as 12.26 W/m2K. Using these values we design for the hotspot.

8.1.7 Accessories design

8.1.7.1 Nozzles

8.1.7.2 Manhole

20-inch (0.5 meter) manhole must be designed for the reactors with diameter more than one meter. Also, only one manhole is required for the single bed vessels. Therefore, one 0.5 meter manhole will be used.

8.1.7.3 Bursting disc

Bursting discs are safety devices designed and manufactured to relieve the system pressure if the pressure of the vessel exceeds the predetermined value. Generally, minor pressure increments can be compensated by the relief valves, however, if the system pressure overruns the 20% of maximal design pressure, then bursting discs fail and provide full hole opening for flow. Depending on the corrosivity of the fluid, variety of construction material for the bursting discs can be used. The size range of bursting disks is from 3 mm to 1.2 m and diameter of the rupture disk can be assumed as 0.5 meter .

8.1.7.4 Wind Loads and Dynamic Wind Pressure

The load and speed of the wind should be considered when designing equipment for process plants. Therefore, the meteorological factors need to be analyzed elaborately for both onshore and offshore operations. Pressure that the wind impact to cylindrical vessels can be found by the equation below:

$wP_W = 0.05 * u^2$ (2.8)

In which,

P_w − wind pressure (P),

u_w − wind speed (km/hr).

The wind velocity depends on the local topography and some other factors of the chosen location. Hai Pong city of Vietnam is selected as the target location for the MMA plant and maximal velocity of wind in Hai Pong is recorded as 15 km/hr in January which is the windiest month in Hai Pong [40]. However, calculations should be done for 30 km/hr for safer design.

$P_w = 0.05 * 30^2 = 45\ Pa$

Now, the load imposed per reactor length can be evaluated:

$$W = P_w * D_{eff} \quad (2.9)$$

W − load per unit length (N/m),

D_{eff} − effective diameter (m).

Effective diameter is calculated as the sum of reactor outside and insulation diameters, however, for isothermal reactors insulation is not installed and therefore, can be skipped.

$W = P_w * D_{eff} = 45 * 3.265 = 147\ N/m$

Knowing the load per unit length, it is feasible to determine the bending moment:

M_x 2

$(W * H$ $)=$ 12.86^2 2 $= 12152.6\ N * m$

$\underline{vessel}$ $= 147$

$* ^2$

M_x − maximal bending moment (N*m),

H_{vessel} − overall height of vessel (m).

Weight of Vessel

Approximate weight of the multitubular isothermal steel fixed bed reactor can be calculated by knowing the important parameters [37]:

$$W_V = 240 * C_V * D_m * (H_{vessel} + 0.8 * D_m) * t_{wall}$$

Where,

W_V − reactor weight excluding internals and fittings(kN),

C_V − vessel internals constant accounts for weight of internals (taken as 1.15),

D_m − effective mean diameter, which is the sum of internal diameter and wall thickness (m).

$$D_m = D_{internal} + t_{wall} = 3.24 + 0.014 = 3.25\ m$$

$$W_V = 240 * 1.15 * 3.25 * (12.86 + 0.8 * 3.25) * 0.014 = 194.5\ kN$$

8.2 Extraction column

The step-by-step design calculations for a pulsed sieve-plate liquid-liquid extraction (LLE) column is given in this section. The column is designed to recover Methacrylic Acid (MAA) from an aqueous process stream using n-Butyl Acetate (n-BuAc) as the extraction solvent. Distribution coefficients for all components were obtained from UNIFAC activity-coefficient simulation. (Attached in annexure 3)

The design methodology follows the standard Kremser equation approach for multi-stage extraction, combined with empirical correlations for column sizing (Thornton flooding correlation) and mass-transfer efficiency (Height Equivalent to a Theoretical Stage, HETS).

8.2.1 Feed Stream Characterisation

The aqueous feed stream entering the extraction column is the liquid outlet of the crystallizer C203. The feed composition is given in Table 30: Inlet to the extraction column.

Table 30: Inlet to the extraction column

Component	Mol %	Molar flow rate (kmol/h)	Flow rate (kg/h)	Wt %
MAA (Methacrylic acid)	10.55	24.70	2,124	35.0
Methacrolein (MAC)	0.07	0.17	12	0.2
Acrylic acid	0.14	0.32	23	0.4
Acetic acid	1.43	3.36	201	3.3
Water	87.81	205.64	3,701	61.1
Total	100.00	234.19	6,061	100.0

The bulk density of the aqueous feed mixture is estimated as a weighted average: $\rho \approx 1013$ kg/m^3, giving a volumetric feed flow of 1.663×10^{-3} m^3/s (or 5.98 m^3/h).

8.2.3 Solvent Selection & Distribution Coefficients

Butyl Acetate (n-BuAc) is selected as the extraction solvent on the basis of:

1. High selectivity for methacrylic acid over water (K_D = 8 >> 1).
2. Low mutual solubility with water (~0.7 wt%), minimising solvent contamination of the raffinate.
3. Readily recoverable by distillation (b.p. 126 °C), enabling economical solvent recycle.
4. Chemically inert toward MAA and MAC under extraction conditions.

8.2.4 UNIFAC Distribution Coefficients

The distribution coefficient K_D is defined as the equilibrium concentration ratio of a component in the organic (extract) phase to the aqueous (raffinate) phase:

$$K_D = \frac{c_{org}}{c_{aq}} = \frac{(mass\, fraction\, organic)}{(mass\, fraction\, aqueous)}$$

Values were obtained from UNIFAC group-contribution activity-coefficient simulation at process temperature (13 °C) are given in Table 31: Distribution coefficients using UNIFAC

Table 31: Distribution coefficients using UNIFAC

Component	K_D	Affinity for n-BuAc	Design Implication
MAA (Methacrylic acid)	8	High	Primary target; good extraction driving force
Methacrolein (MAC)	15	Very high	Near-complete co-extraction

Component	K_D	Affinity for n-BuAc	Design Implication
			expected
Water	0.35	Low	Remains predominantly in aqueous raffinate
Acetic acid	1.0	Moderate	Partial co-extraction (~19%)
Acrylic acid	2.5	Moderate	Significant co-extraction (~73%)
n-Butyl Acetate (solvent)	850	Stays in organic	Negligible loss to raffinate

There is significant co-extraction of acrylic acid into the organic phase means that downstream MAA purification (e.g. azeotropic or extractive distillation) must achieve MAA / acrylic acid separation.

8.2.5. Solvent-to-Feed Ratio

8.2.5.1 Minimum Solvent Ratio

The minimum solvent-to-feed mass ratio is derived from single-stage equilibrium theory. When the extraction factor E equals exactly 1.0, the minimum solvent is just sufficient to carry the solute at thermodynamic equilibrium.

$$(S/F)\_min = 1 / K\_D(MAA) = 1 / \ln(8) = 0.455 \quad (mole\ basis)$$

which is approximately 2.2 in mass basis.

8.2.5.2 Design Solvent Ratio

A safety factor of 1.5× is applied to the minimum to ensure a feasible extraction factor (E > 1) and to accommodate uncertainty in the UNIFAC K_D estimate and feed composition variability-

$$S/F = 1.5 \times (S/F)\_min = 1.5 \times 2.2 = 3.3$$

Solvent flow = 3.3 × 6,061 kg/h = 20,001.3 kg/h (22.68 m^3/h)

8.2.6. Stage Calculation — Kremser Equation

8.2.6.1 Extraction Factor

The extraction factor E characterises the thermodynamic driving force for transfer from the aqueous to the organic phase -

$$E = K_D \times (S/F) = 8 \times 3.3 = 26.4$$

Since E > 1, extraction is feasible and the required number of stages is finite. This is a high enough to achieve the target recovery in a reasonable number of stages without excessive solvent use.

8.2.6.2 Kremser Equation for Number of Theoretical Stages

The Kremser equation relates the number of theoretical equilibrium stages N_{th} to the fractional recovery f and extraction factor E-

$$N_{th} = \ln[(1 - 1/E) \times (x_{in} / x_{out}) + 1/E] / \ln(E)$$

Where $x_{in} / x_{out} = 1 / (1 - f)$. Substituting the design values (f = 0.97, E = 26.4) we get 1.6 theoretical stages.

8.2.6.3 Overall Stage Efficiency

Real stages are less effective than ideal equilibrium stages due to incomplete phase contact, droplet coalescence, and axial dispersion. The overall stage efficiency E_o for a pulsed sieve-plate column operating with an organic acid / ester solvent system is typically in the range 0.40–0.55. A conservative value of 0.45 is adopted. $E_o = 0.45$ (overall stage efficiency)

8.2.6.4 Number of Actual stages

$$N_{act} = N_{th} / E_o = 1.07 / 0.45 = 3.5, \text{ I.e 4 actual stages.}$$

The plate spacing is set at the standard value of 50 mm for pulsed sieve-plate columns.

8.2.6.5 Multi-Component Kremser Analysis

The same Kremser framework is applied to each component using its respective K_D value and the design S/F ratio.

Table 32: Kremser analysis for multicomponent

Component	K_D	E factor	N_th (97%)	N_act needed	Recovery at N=6
MAA	8	26.1	1.0	4	97.0%
Methacrolein	15	32.820	1.0	4	98.4%
Acrylic acid	2.5	0.470	N/A (E<1)	∞	73.1%
Acetic acid	1.0	0.188	N/A (E<1)	∞	19.3%
Water	0.35	0.066	N/A (E<1)	∞	6.8%

Components with $E < 1$ (acrylic acid, acetic acid, water) cannot achieve 97% recovery regardless of stage count meaning that their extraction is thermodynamically limited. MAA governs the stage count as the primary target component with the lowest distribution coefficient among desirable extracts.

8.2.7. Column Diameter

8.2.7.1 Flooding Velocity

Column diameter is determined by limiting the superficial velocity to a safe fraction of the flooding velocity. The Thornton correlation for pulsed sieve-plate columns with a low-interfacial-tension solvent pair (n-BuAc / water, $\sigma \approx 14$ mN/m) gives the flooding velocity as ≈ 0.018 m/s (Thornton correlation, pulsed column, $\sigma = 14$ mN/m).

8.2.7.2 Design Velocity and Cross-Section

A design velocity of 60% of flooding is standard practice for pulsed columns, providing operational flexibility and avoiding premature phase inversion:

$$u\_design = 0.60 \times u\_flood = 0.60 \times 0.018 = 0.0108 \ m/s$$

$$Q\_total = Q\_feed + Q\_solvent = 0.008 \ m^3/s$$

$$A\_req = Q\_total / u\_design = 0.008/ 0.0108 = 0.737 \ m^2$$

8.2.7.3 Selected Diameter

The diameter of the column can be calculated by

$$D = (4A\_Req/3.14)^{0.5}$$

This formula gives diameter as 0.969 m, which will be rounded off to 1m for standard sizing. The selected diameter of 1 m gives an actual operating velocity of **~0.0104 m/s**, which is approximately 58% of flooding — well within the recommended operating window.

8.2.8. Column Height

8.2.8.1 Active Height (HETS Method)

The active extraction height is calculated from the number of theoretical stages and the Height Equivalent to a Theoretical Stage (HETS). For pulsed sieve-plate columns operating with organic-acid systems, a representative HETS value is 0.25 m/stage, consistent with literature data for similar systems-

$$H\_active = N \times HETS = 4 \times 0.25 = 1.52 \ m$$

8.2.8.2 Total Column Height

Additional height is added for the phase disengagement sections at the top and bottom of the column, plus nozzle/pulse-leg allowances. The height of the column will be 2.7 m.

Table 33: Height calculations for the extraction column

Section	Height (m)	Basis
Active extraction zone	1.00	N × HETS
Top settler (organic disengagement)	0.80	Standard allowance
Bottom settler (aqueous disengagement)	0.60	Standard allowance
Nozzle / pulse-leg allowance	0.30	Piping connection
Total column height	2.7	

8.2.9 Pulsation Conditions

The column operates as a pulsed sieve-plate extractor, where mechanical pulsation promotes drop breakup and enhances interfacial mass transfer. The pulsation intensity is characterised by the product of amplitude A and frequency f (the Af product).

Table 34: Pulsating frequencies

Parameter	Recommended Value	Basis
Pulse amplitude A	8 – 12 mm	n-BuAc/water, σ = 14 mN/m
Pulse frequency f	1.0 – 1.5 Hz	Emulsion regime target
Af intensity	10 – 15 mm/s	Optimal drop size for K_D = 8

Plate free area	20 – 23%	Standard pulsed sieve tray
Hole diameter d_h	3 mm	Avoids weeping at low Af
Plate spacing	50 mm	Standard; matches HETS estimate
Dispersed phase	Organic (n-BuAc)	Lower flow rate → dispersed
Material of construction	SS 316L	Methacrylic / acrylic acid service

Operating regime: Pulsed columns exhibit three operating regimes: mixer-settler (Af < 8 mm/s), emulsion (Af = 8–18 mm/s), and flooding (Af > 20 mm/s). This design targets the emulsion regime for maximum mass-transfer efficiency. Operating below 8 mm/s risks poor mass transfer; above 20 mm/s risks phase inversion and flooding.

8.2.10. Stream Table

Table 35: Mass flow rates of the streams for the LLE column

Component	Feed (Aqueous)	Solvent In (n-BuAc)	Extract (Organic out)	Raffinate (Aqueous out)
MAA (Methacrylic acid)	2,124.0	0.0	2,066.7	57.3
Methacrolein (MAC)	12.0	0.0	12.0	0.0

Acrylic acid	23.0	0.0	22.9	0.1
Acetic acid	201.0	0.0	201.0	0.0
Water	3,701.0	0.0	986.2	2,714.8
n-Butyl Acetate (solvent)	0.0	20,001.3	18,568.0	1,433.3
Total mass flow (kg/h)	6,061.0	20,001.3	21,856.8	4,205.5

8.3 Distillation columns

This section has the sizing of three distillation columns required to purify Methacrylic Acid (MAA) from the downstream of a liquid–liquid extractor. The extraction solvent is n-Butyl Acetate (nBA). The design follows the Fenske–Underwood–Gilliland (FUG) method, a well-established shortcut procedure for column sizing at the conceptual and pre-FEED stage.

Objective: Recover ≥ 1,900 kg/h of MAA at ≥ 99.5 wt% purity from the extractor effluent, while recovering substantially all nBA for recycle to the extraction unit.

8.3.1 Feed Composition

The feed to the distillation train is the extract-phase effluent from the liquid–liquid extractor. The composition is given in Table 36: Feed composition for the distillation column.

Table 36: Feed composition for the distillation column

Component	**Flow rate (kg/h)**	**Wt%**	**Type**
MAA (Methacrylic acid)	2,066.7	9.45%	Key product
Methacrolein	12.0	0.05%	Light impurity
Acrylic acid	22.9	0.10%	Heavy impurity
Acetic acid	201.0	0.92%	Heavy impurity
Water	986.2	4.51%	Co-extracted
n-Butyl acetate (nBA)	18,568	84.95%	Extraction solvent
TOTAL	21,856.8	100%	

8.3.2 Separation Strategy — Three-Column Train

The mixture contains components spanning a wide boiling-point range. Three columns are required due to the presence of both a light solvent (nBA) and closely-boiling acid impurities that must be separated from MAA at high purity. MAA is required to be at high purity since it has to be reacted further with ethylene oxide to form HEMA. The proposed separation sequence is as follows-

1. Column 1 (C1): Atmospheric pressure — Remove nBA and methacrolein overhead; send concentrated organics + water to C2.
2. Column 2 (C2): Vacuum (45 kPa) — Recover purified nBA as distillate for solvent recycle; send MAA-rich bottoms to C3.
3. Column 3 (C3): High vacuum (12 kPa) — Separate light impurities (acetic acid, acrylic acid, water) overhead and recover ≥99.5 wt% MAA as bottoms product.

Operating C2 and C3 under vacuum reduces the boiling points of MAA and the acids, keeping reboiler temperatures within a safe operating window (below ~80–90 °C) to prevent polymerisation.

8.3.2 Design Basis & Assumptions

Table 37: Summary of the key assumptions

Parameter	**Value**

Design method	FUG shortcut
Tray type	Sieve trays
Tray spacing	500 mm
Murphree efficiency	60%
Reflux multiplier	$1.2 \times R_{min}$
Flooding fraction	80%
F-factor limit	1.0 $Pa^{0.5}$
Downcomer area	12% of total
Feed condition	q = 1 (sat. liq.)
Top space allowance	1.5 m
Bottom sump allowance	1.5 m

8.3.3 Design Method & Formulas

The Fenske–Underwood–Gilliland (FUG) shortcut method is used throughout. This section presents all governing equations in the order applied.

8.3.3.1 Fenske Equation — Minimum Theoretical Stages (N_min)

The Fenske equation gives the minimum number of theoretical stages at total reflux ($R \rightarrow \infty$):

$$N_{min} = \log[(x_{LK}/x_{HK})_D \times (x_{HK}/x_{LK})_B] \;/\; \log(\alpha_{LK,HK})$$

Where:

- x_{LK}, x_{HK} = mole fractions of light key (LK) and heavy key (HK) in distillate (D) and bottoms (B)
- $\alpha_{LK,HK}$ = relative volatility of LK with respect to HK, evaluated at average column temperature and pressure
- N_{min} is the number of theoretical equilibrium stages at total reflux (lower bound)

8.3.3.2 Underwood Equation — Minimum Reflux Ratio (R_min)

The Underwood method finds R_{min} by solving for θ (the Underwood root) and then summing over all components-

Step 1 — Find θ satisfying (one root between α_{LK} and α_{HK}):

$$\Sigma \left[\alpha_i \times z_i / (\alpha_i - \theta) \right] = 1 - q$$

Step 2 — Calculate R_{min}:

$$R_{min} + 1 = \Sigma \left[\alpha_i \times x_{D,i} / (\alpha_i - \theta) \right]$$

Where:

- z_i = feed mole fraction of component i
- q = feed thermal condition (q = 1 for saturated liquid; q = 0 for saturated vapour)
- $x_{D,i}$ = distillate mole fraction of component i
- θ = Underwood root (solved iteratively; lies between $\alpha_{HK} < \theta < \alpha_{LK}$)

8.3.3.3 Gilliland Correlation — Actual Theoretical Stages (N_th)

The Gilliland correlation relates the actual number of stages to N_{min} and R_{min} through the empirical Molokanov equation:

$$X = (R - R_{min}) / (R + 1)$$

$$Y = 1 - \exp\left[(1 + 54.4 \cdot X) / (11 + 117.2 \cdot X) \times (X - 1) / X^{0.5} \right]$$

$$N_{th} = (N_{min} + Y) / (1 - Y) \quad \textit{[rearranged from Gilliland chart]}$$

Operating reflux ratio:

$$R_{op} = 1.2 \times R_{min}$$

The factor 1.2 is a conservative choice. Optimal R_{op} minimises total annual cost (TAC) and is typically 1.1–1.5 × R_{min} for most organic systems.

8.3.3.4 Tray Efficiency — Actual Number of Trays

The Murphree vapour-phase efficiency (E_{mv}) relates actual to theoretical stages:

$$N_{actual} = N_{th} / E_{mv}$$

$$E_{mv} = 0.60 \quad \textit{(60\%, assumed for sieve trays in organics + water systems)}$$

8.3.3.5 Optimal Feed Tray Location (Kirkbride Equation)

The Kirkbride equation estimates the optimal feed stage location:

$$\log(N_r / N_s) = 0.206 \times \log[(B/D) \times (z_{HK}/z_{LK}) \times (x_{LK,B}/x_{HK,D})^2]$$

Where N_r = number of stages above feed (rectifying), N_s = below feed (stripping), B = bottoms flow, D = distillate flow.

8.3.3.6 Column Diameter — F-Factor (Souders–Brown) Method

The column diameter is sized to keep the vapour velocity below the flooding velocity, using the F-factor (Souders–Brown coefficient):

$$F = u_v \times \sqrt{\rho_v} \qquad [Pa^{0.5} \text{ or } m/s \times (kg/m^3)^{0.5}]$$

At flood: $F_{flood} = C_{sb} \times \sqrt{[(\rho_L - \rho_v) / \rho_v]}$

Design at 80% flood: $u_{v,design} = 0.80 \times u_{v,flood}$

Net area required: $A_{net} = V_{mass} / (F_{limit} \times \sqrt{\rho_v})$

$$F_{limit} = 1.0\ Pa^{0.5} \text{ (sieve trays, 500 mm spacing)}$$

Total cross-section: $A_{total} = A_{net} / (1 - f_{DC})$

$f_{DC} = 0.12$ (12% downcomer area fraction)

Column diameter: $D_{col} = \sqrt{4 \times A_{total} / \pi}$

Where:

- V_{mass} = vapour mass flow rate at the column top (kg/s), from material balance
- ρ_v = vapour density at operating temperature and pressure (kg/m^3), estimated from ideal gas law
- ρ_L = liquid density (kg/m^3)

8.3.3.7 Column Height

The total column height is calculated as:

$$H_{trays} = N_{actual} \times tray\_spacing$$

$$= N_{actual} \times 0.500 \ [m]$$

$$H_{total} = H_{trays} + H_{top} + H_{bottom}$$

$$= H_{trays} + 1.5 + 1.5 \ [m]$$

8.3.4. Column-by-Column Sizing Results

8.3.4.1 Column 1 - nBA / Organics Splitter, T302

Pressure: 101.3 kPa (atmospheric)   Function: Remove nBA overhead; send concentrated organics + water to C2

Table 38: Summary of design of column T302

Parameter	**Value**	**Units**	**Source Formula**	**Notes**
Pressure	101.3	kPa	—	Atmospheric
LK / HK	AcOH / MAA	—	—	Acetic acid / MAA
Rel. volatility α	3.8	—	Antoine eqns at avg T, P	AcOH/ MAA at 1 atm
R_min (Underwood)	0.85	—	$\Sigma[\alpha \cdot z/(\alpha-\theta)] = 1-q \rightarrow R_{min}$	
R_op = 1.2 × R_min	1.02	—	R_op = 1.2 × 0.85	

N_{min} (Fenske)	6	stages	Fenske eq.	At total reflux
$N_{theoretical}$ (FUG)	14	stages	Gilliland / Molokanov	At R_{op}
N_{actual} (60% eff.)	24	trays	$N_{th} / 0.60 = 23.3 \rightarrow 24$	Rounded up
Feed tray (from top)	12	—	Kirkbride eq.	
Vapour flow at top	4.8	kg/s	Material balance + reflux	
Vapour density at top	3.2	kg/m^3	Ideal gas at top T, P	
$A_{net} = V/(F\cdot\sqrt{\rho_v})$	2.68	m^2	$4.8 / (1.0 \times \sqrt{3.2})$	
$A_{total} = A_{net}/(1-0.12)$	3.04	m^2	2.68 / 0.88	
Column diameter	1.8	m	$\sqrt{(4\times3.04/\pi)} \rightarrow$ rounded	Std. shell size

Tray section height	12.0	m	24 × 0.500	
Top + bottom allowance	3.0	m	1.5 + 1.5	
TOTAL COLUMN HEIGHT	~17.4	m	12.0 + 3.0 + 2.4 (nozzles)	incl. nozzle allow.

8.3.4.2. Column 2 - nBA Recovery Column, T303

Pressure: 45 kPa (vacuum)   Function: Recover purified nBA as distillate for recycle; bottoms to C3

Table 39: Summary of design of column T303

Parameter	**Value**	**Units**	**Source Formula**	**Notes**
Pressure	45.0	kPa	—	Vacuum
LK / HK	nBA / AcOH	—	—	

Rel. volatility α	5.2	—	Antoine eqns at avg T, P	nBA/AcOH at 45 kPa
R_min (Underwood)	0.60	—	Underwood method	
R_op = 1.2 × R_min	0.72	—	R_op = 1.2 × 0.60	
N_min (Fenske)	5	stages	Fenske eq.	
N_theoretical (FUG)	12	stages	Gilliland / Molokanov	
N_actual (60% eff.)	20	trays	N_th / 0.60 = 20	
Feed tray (from top)	10	—	Kirkbride eq.	
Vapour flow at top	5.9	kg/s	Material balance	Dominated by nBA
Vapour density at top	2.1	kg/m³	Ideal gas at 45 kPa	

$A_{net} = V/(F \cdot \sqrt{\rho_v})$	4.07	m^2	$5.9 / (1.0 \times \sqrt{2.1})$	
$A_{total} = A_{net}/0.88$	4.63	m^2	4.07 / 0.88	
Column diameter	2.2	m	$\sqrt{(4 \times 4.63/\pi)} \rightarrow 2.43 \rightarrow 2.2$	Std. shell size
Tray section height	10.0	m	20×0.500	
Top + bottom allowance	3.0	m	1.5 + 1.5	
TOTAL COLUMN HEIGHT	~15.0	m	10.0 + 3.0 + 2.0	

8.3.5.3 Column 3 - MAA Purification Column, T304

Pressure: 12 kPa (high vacuum)   Function: Remove light acids overhead; recover ≥99.5 wt% MAA as bottoms product

Table 40: Summary of design of column T304

Parameter	Value	Units	Source Formula	Notes
Pressure	12.0	kPa	—	High vacuum
LK / HK	AcOH / MAA	—	—	
Rel. volatility α	6.1	—	Antoine eqns at avg T, P	AcOH/ MAA at 12 kPa
R_min (Underwood)	1.10	—	Underwood method	Higher due to purity spec
R_op = 1.2 × R_min	1.32	—	R_op = 1.2 × 1.10	
N_min (Fenske)	8	stages	Fenske eq.	
N_theoretical (FUG)	18	stages	Gilliland / Molokanov	High N from purity spec

N_actual (60% eff.)	30	trays	N_th / 0.60 = 30	
Feed tray (from top)	18	—	Kirkbride eq.	
Vapour flow at top	0.78	kg/s	Material balance	Much smaller than C1/C2
Vapour density at top	0.85	kg/m^3	Ideal gas at 12 kPa	
$A_{net} = V/(F \cdot \sqrt{\rho_v})$	0.847	m^2	$0.78 / (1.0 \times \sqrt{0.85})$	
$A_{total} = A_{net}/0.88$	0.962	m^2	0.847 / 0.88	
Column diameter	1.2	m	$\sqrt{(4 \times 0.962/\pi)} \rightarrow 1.11 \rightarrow 1.2$	Std. shell size
Tray section height	15.0	m	30×0.500	
Top + bottom allowance	3.0	m	1.5 + 1.5	

TOTAL COLUMN HEIGHT	~21.0	m	15.0 + 3.0 + 3.0	incl. nozzle allowance

8.2.4 Product Streams & Mass Balance

Table 41: Mass flow rates of the streams

Component	**Feed (C1)**	**T302 Dist.**	**T302 Bot.**	**T303 Dist. (nBA)**	**T303 Bot.**	**T304 Dist.**
MAA	2066.7	2.1	2064.6	0.0	2064.6	85.0
Methacrolein	12.0	12.0	0.0	12.0	0.0	0.0
Acrylic acid	22.9	0.5	22.4	0.0	22.4	22.1
Acetic acid	201.0	4.0	197.0	0.0	197.0	195.1
Water	986.2	49.3	936.9	0.0	936.9	936.6

nBA	18568	18512	56.0	18327	229.0	0.0
TOTAL	21856.8	18579.9	3277.0	18339	3449.9	1238.8

The bottoms of the column T304 (MAA Product) wil contain ~1,980 kg/h MAA and purity of 99.5%.

8.2.5 Process Safety and MAA Polymerisation

MAA undergoes free-radical exothermic polymerisation above ~80–90 °C. This must be actively suppressed throughout the distillation train.

1) The following measures are required:
2) Polymerisation inhibitor (MEHQ or hydroquinone) at 300–500 ppm in reboiler of C3; 50–100 ppm in overhead/condenser of C3.
3) If MEHQ is used, a small O_2 co-feed (air purge) is necessary to activate the inhibitor — ensure this does not create a flammable mixture.
4) Thermosiphon reboiler preferred over kettle reboiler for C3 to minimise liquid residence time at elevated temperature.
5) At 12 kPa, the C3 base temperature is approximately 75–80 °C — within safe operating window. Verify this does not creep up due to fouling or pressure control issues\
6) Avoid hot liquid hold-up in the reboiler; ensure no dead zones or stagnant pockets.
7) In C1 at atmospheric pressure, the reboiler temperature is approximately 130 °C — MAA losses may be significant; confirm with rigorous simulation.

9. Hazard and Operability study (HAZOP)

Hazard and Operability (HAZOP) analysis is a qualitative and structured method used to identify possible hazards and accident scenarios in the workplace. The process is broken down into stages, where guide words are applied to examine the consequences of deviations from intended operating conditions. HAZOP is among the most commonly used techniques for process hazard analysis in the chemical industry. It follows a systematic brainstorming approach aimed at detecting and reducing potential risks by analyzing unexpected variations in a process. This method is especially useful in identifying human-related factors that may lead to system failures, as it involves carefully examining deviations from normal operations and evaluating their impact on safety.

The procedure starts by defining the normal operating conditions or the intended function of a system, known as the "aim." Guide words are then used to explore possible deviations from this intended state. When a deviation is identified, its likely causes are recorded, and its potential impact on safety is assessed. Based on these findings, suitable measures are implemented to reduce or eliminate the associated risks.

HAZOP studies focus on identifying potential problems under normal operating conditions, ensuring that each issue is assigned to a responsible person for corrective action. The table below outlines commonly used HAZOP guide words along with their meanings.

Table 42: HAZOP guide words and their meaning, Turton et. Al., 2018

Guide word	**Meaning**
No	Negation of the design intent
Less than	Quantitative decrease
More than	Quantitative increase
Part of	Qualitative decrease
As well as	Qualitative increase

Guide word	Meaning
Reverse	Logical opposite of the intent
Other than	Complete substitution

For this report, reactor R-201, is chosen for the HAZOP study. The nodes selected are the feed entering the reactor, the heat transfer medium, reactor vessel and the catalyst.

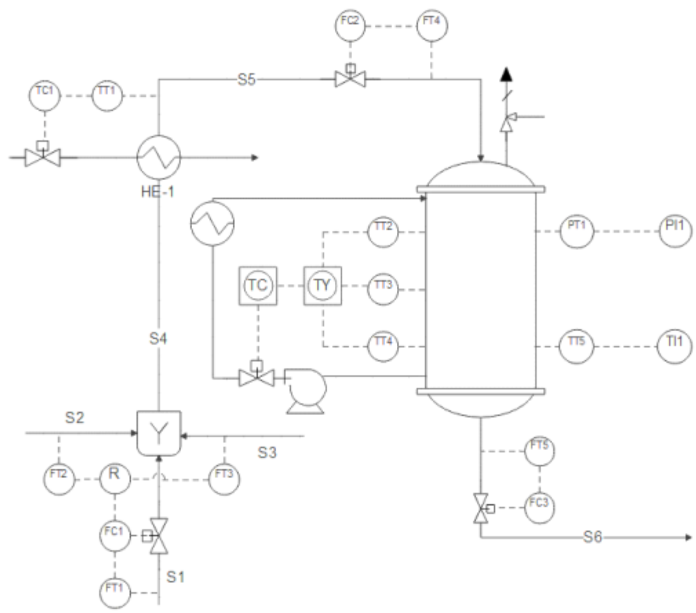


Figure 6: P&ID of the reactor

9.1 Node 1: Feed

The feed is a mixture of isobutene, air and steam. The ratio of isobutene to air to steam is to be strictly maintained as 1: 0.5-5: 1-5, to prevent keep isobutene within the flammability limits. The feed temperature should be maintained to 350 °C to have higher rates and selectivity towards methacrolein.

9.1.1 Composition of feed

Table 43: HAZOP for the feed composition

Parameter	**Guide word**	**Meaning**	**Causes**	**Consequences**	**Actions required**	**Safe guards**
Composition	More	Higher ratio of isobutene to oxygen in the feed	Isobutene flow control valve fails open. Air/oxygen blower fails or is blocked. Oxygen analyzer gives false low reading.	**Entry into flammable range** (risk of explosion in reactor)	Ensure the **Flow Control Valve (FCV)** is set correctly and the oxygen analyser is always calibrated	High isobutene and Low oxygen alarms Ratio indicator on the DCS panel Manual ratio check by the operator
Composition	Less	Lesser ratio of isobutene to oxygen	Isobutene feed flow controller fails partially closed. Oxygen/air blower	Over oxidation of isobutene to carbon	Implement cascade ratio control (isobutene	Low isobutene and High oxygen flow

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
		in the feed	surges or overspeeds. Partial blockage in isobutene feed line or filter. Gas analyzer gives false high isobutene reading → controller reduces isobutene unnecessarily.	dioxide and carbon monoxide. **Entry into flammable range** (risk of explosion in reactor	flow setpoint derived from measured oxygen flow). Install an online gas analyzer (GC or IR) for isobutene and oxygen with low ratio alarm at operator panel. Regular calibration of flow meters and analyzers.	alarm (existing). Ratio indicator on DCS panel. Manual daily ratio check by operator.

9.1.2 Temperature

Table 44: HAZOP for the feed temperature

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
Temperature	More	The temperature of the feed entering the reactor is more than 350 °C	Feed preheater temperature controller fails (valve stuck open). Heat exchanger fouling on the heat transfer medium (cooling) side. Solar heating of feed lines due to fault in the insulation	Increased risk of **premature ignition** or auto-ignition of isobutene/air mixture before reaching catalyst. **Flammability range widens** the lower explosive limit (LEL) decreases. Potential for **runaway reaction starting in the feed line** and fire hazard	Install high feed temperature alarm with automatic feed diversion or shutoff. Ensure preheater control valve fails closed (air-to-open configuration). Provide temperature monitoring at reactor inlet nozzle (not just preheater outlet). Regularly calibrate	High temperature alarm on feed line. Temperature controller with interlock. Manual operator check of feed temperature per shift. Line insulation review

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
					temperature sensors.	
Temperature	Less	The temperature of the feed entering the reactor is less than 350 °C	Preheater failure (steam/electrical off). Heat exchanger bypass valve left open. Ambient cold conditions (winter) with insufficient tracing. Temperature controller fails (valve stuck closed). Cold feed from storage (if no preheater).	Possible condensation of isobutene or water in feed line Liquid in catalytic reactor can cause catalyst cracking or pressure surges. Slow reaction initiation at catalyst bed. Poor conversion and low methacrolein yield. May require	Install low feed temperature alarm. Ensure preheater control valve fails open (air-to-close) for steam heating, or has a minimum stop. Provide steam tracing or electric heat tracing on feed lines in cold climates. Regular maintenance	Low temperature alarm. Temperature indicator on DCS. Operator procedure to check feed temperature before startup.

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
				more oxygen to compensate	of preheater and heat exchanger.	

9.1.3 Pressure

Table 45: HAZOP for the feed pressure

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
Pressure	More	The pressure of the feed entering the reactor is more than 2.2 bar abs	Pressure control valve fails closed downstream of feed blower/compressor Compressor surge or overspeed. Operator error in setting pressure regulator. Check valve stuck closed	Mechanical damage to reactor inlet nozzle or catalyst bed support grid. Leak or rupture of feed piping (isobutene/air is flammable and potentially explosive).	Install high feed pressure alarm with automatic compressor trip or feed shutoff. Install a **pressure relief valve** on the feed line upstream of the reactor	High pressure alarm (PIASH). Pressure relief valve (PSV) vented to safe location. Compressor anti-surge control. Rupture disc

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
			downstream.	Increased risk of backfire from reactor into feed line if pressure differential reverses. Potential damage to flow meters (especially thermal mass flow meters).	inlet. Ensure pressure control valve fails open (air-to-close) or has a bypass. Regular inspection of feed line filters. Use a **flame arrestor** between feed line and reactor to prevent backfire propagation.	on feed line (if code requires). Regular filter cleaning schedule.
Pressure	Less	The pressure of the feed entering the reactor is	Feed compressor or blower failure. Suction filter blockage. Leak in feed	Potential for **oxygen backflow** from reactor into feed line if	Install low feed pressure alarm with automatic trip or feed	Low pressure alarm (PIAL). Check valve on feed line.

Parameter	**Guide word**	**Meaning**	**Causes**	**Consequences**	**Actions required**	**Safe guards**
		less than 2.2 bar abs	piping upstream (loss of containment). Pressure control valve fails open. Operator reduces pressure setpoint incorrectly.	pressure drops below reactor pressure (requires check valve). Low conversion of the feed	diversion. Install a **check valve** on feed line to prevent backflow from reactor. Provide a pressure transmitter at reactor inlet (not just at compressor discharge). Regular compressor maintenance.	Flow monitoring (low flow alarm indirectly indicates low pressure). Compressor suction pressure monitoring. Operator training on leak response.

9.2 Node 2: Heat transfer fluid (Thermic salt)

For cooling the reactor, a thermic salt ($LiNaKNO_3$) is used. The coolant flows through the shell side and the parameters that may affect the safety and operability of the reactor.

9.2.1 Temperature

Table 46: HAZOP for the thermic salt temperature

Parameter	**Guide word**	**Meaning**	**Causes**	**Consequences**	**Actions required**	**Safe guards**
Temperature	More	Thermic salt inlet temperature is higher than the normal operating range of 460 K	Cooling system for the salt fails such as an air cooler or fin fan trip Recirculation pump provides insufficient flow causing stagnation and local heating Exothermic reaction inside the reactor tubes is too strong transferring more heat to the salt than normal Hot spots in the reactor due to uneven catalyst loading or feed maldistribution Temperature controller sends false low signal	Reduced heat removal capacity from the reactor tubes because the temperature driving force decreases Increased risk of hot spots inside the catalyst bed Catalyst sintering or permanent deactivation Runaway exothermic reaction if heat cannot	Install high temperature alarm on thermic salt inlet with automatic action Implement an automatic trip that cuts isobutene feed if salt inlet temperature exceeds a setpoint Provide a second independent temperature sensor for interlock purposes	High temperature alarm on DCS Independent temperature interlock that trips isobutene feed Regular salt sampling and analysis schedule Operator training on runaway reaction symptoms Emergency response procedure for high salt temperature

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
			causing heater to run continuously	be removed fast enough Thermic salt degradation such as nitrite oxidizing to nitrate which changes thermal properties Increased viscosity of the salt leading to poor pumping and further temperature rise	Install a bypass cooler or emergency cooling system for the salt Regularly analyze salt composition to monitor degradation	
Temperature	Less	Thermic salt inlet temperature is lower	Thermic salt heater or heat exchanger control valve fails closed	Poor reaction initiation at the reactor	Install low temperature alarm on thermic salt	Low temperature alarm on DCS Heater

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
		than the normal operating range of 460 K	Steam supply to the salt heater fails if steam heated Circulation pump trips causing cold salt to enter from the expansion tank or standby system Ambient cooling of salt lines during low flow conditions Temperature controller fails sending false high signal causing heater to turn off Cold makeup salt is added too quickly without preheating	inlet because the catalyst is too cold Reduced conversion of isobutene leading to low MAA yield Increased byproduct formation such as carbon monoxide or carbon dioxide due to incomplete oxidation Risk of salt freezing or solidification if temperature	inlet Ensure heater control valve fails open or has a minimum stop for steam heated systems Implement a low temperature interlock that cuts isobutene feed or diverts salt flow Provide steam tracing or electric heat tracing on salt lines in cold	control valve with fail open position Steam tracing on critical salt lines Pump motor current monitoring Pre startup salt melting checklist

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
				drops below the melting point of 451 K Solidified salt can block tubes and damage the reactor or burst piping High viscosity increases pump power and may cause cavitation	climates Establish a salt melting procedure before startup Monitor pump motor current as an indicator of viscosity changes	

9.2.2 Flow rate

Table 47: HAZOP for the thermic salt flow rate

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards

Flow rate	More	Thermic salt flow rate is higher than the required operating range	Thermic salt circulation pump operates at higher speed due to variable frequency drive failure or operator error Pressure control valve on the salt line fails open or is manually opened too far Restriction downstream is removed such as a filter being cleaned or a bypass opened Multiple pumps are running in	Increased cooling capacity which may over cool the reactor tubes Lower reactor temperature leading to poor conversion of isobutene Reduced Methacrolein yield because the reaction is too cold Higher pump power consumption and energy cost Increased erosion of salt piping especially at elbows and tube inlets	Install high flow alarm on thermic salt circulation Implement a maximum flow limit in the variable frequency drive or pump controller Provide a flow control valve that limits maximum flow regardless of operator setpoint Regularly inspect piping for erosion especially at bends Balance pump	High flow alarm on DCS Variable frequency drive with high speed limit Flow control valve with maximum stop Erosion monitoring schedule for piping Pump interlock to prevent unnecessary parallel operation

			parallel when only one is required Viscosity of the salt decreases due to high temperature reducing resistance to flow	Potential vibration in the piping system from high velocity Cavitation on the pump suction side if flow is too high for the net positive suction head available	operation so only required pumps are running	
Flow rate	Less	Thermic salt flow rate is lower than the required operating range	Thermic salt circulation pump fails or trips Variable frequency drive reduces speed due to controller error or sensor failure	Reduced heat removal from the reactor tubes leading to rising tube wall temperatures Formation of hot spots in the	Install low flow alarm with automatic action Implement a low low flow interlock that cuts isobutene feed to the reactor	Low flow alarm on DCS Low low flow interlock that trips isobutene feed Automatic backup pump start Redundant flow

			Partial blockage in the salt lines such as solidified salt debris or corrosion products Filter or strainer becomes blocked with salt degradation products Control valve fails partially closed Air or gas entrapment in the salt circuit causing vapor lock Low salt level in the expansion tank causing pump	catalyst bed Increased risk of runaway exothermic reaction Catalyst deactivation from high temperature exposure Higher salt outlet temperature as the same heat is absorbed by less salt Potential localized boiling or degradation of the salt If flow stops completely the reactor can overheat within minutes	Provide a backup pump that starts automaticall y on low flow or pump failure Install a flow transmitter with two separate elements for reliability Regularly clean salt strainers and filters Monitor pump motor current as an indicator of low flow or cavitation Maintain minimum salt level in the	transmitters Regular strainer cleaning schedule Pump motor current monitoring Low salt level alarm on expansion tank

			cavitation Viscosity increases due to low temperature making the salt thicker		expansion tank	
Flow rate	No	Complete loss of thermic salt circulation	Both primary and backup circulation pumps fail Total loss of electrical power to all pumps Major leak or rupture in the salt piping system Expansion tank runs dry Blockage completely obstructs the salt circuit	Extremely rapid temperature rise in the reactor tubes Runaway exothermic reaction within minutes Severe catalyst sintering and permanent destruction Reactor tube rupture from thermal stress and	Install a flow switch or flow transmitter with a no flow alarm that triggers an immediate isobutene feed trip Provide uninterruptible power supply to the salt pump control system and critical instruments	Flow switch with no flow trip Uninterruptible power supply Diesel driven backup pump Safety instrumented system for loss of flow Regular trip testing schedule Isobutene feed valves that fail closed on loss of

			such as a closed valve or solidified salt plug Operator error isolates the salt circuit during maintenance and forgets to reopen	high pressure Possible explosion of the reactor shell if pressure relief is inadequate Release of toxic and flammable isobutene and MAA Major fire and potential loss of containment Complete plant shutdown and extended repair	Install a backup diesel driven salt pump for emergency circulation Design the system with a high integrity safety instrumented function for loss of flow Perform regular testing of flow switches and trip logic Provide automatic closure of isobutene feed valves on loss of salt flow	power Emergency operating procedures

					Train operators on emergency response to no flow condition	

9.2.3 Pressure

Table 48: HAZOP for the heat transfer fluid pressure

Parameter	**Guide word**	**Meaning**	**Causes**	**Consequences**	**Actions required**	**Safe guards**
Pressure	More	Thermic salt pressure is higher than the normal operating range	Blockage downstream of the pump such as a closed valve or plugged heat exchanger Control valve on the salt line fails closed or is manually closed Check	Mechanical stress on the reactor shell which is not designed for high pressure on the shell side Leak or rupture of salt piping gaskets or expansion joints Damage to	Install high pressure alarm on the thermic salt circuit Provide a pressure relief valve on the salt pump discharge line Ensure control valves fail	High pressure alarm Pressure relief valve vented to a safe salt collection system Control valve with fail open configuration High

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
			valve downstream sticks in the closed position Pump operates at excessive speed Expansion tank pressure control valve fails closed Gas accumulation in the salt circuit expands when heated	the pump seals leading to salt leakage Release of hot molten salt which can cause severe burns and fires Pressure relief valve opening and releasing salt to a collection system Potential reverse flow or damage to the reactor tubes if shell pressure exceeds tube pressure	open or have a bypass for pressure protection Install a pressure transmitter with interlock that trips the salt pump on high pressure Regularly inspect and clean salt filters and strainers Provide a minimum flow bypass line around the pump to prevent dead heading	pressure interlock to trip pump Regular filter inspection schedule Minimum flow bypass line

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
Pressure	Less	Thermic salt pressure is lower than the normal operating range	Leak in the salt piping system such as a cracked weld or failed gasket Pump suction strainer is blocked Low salt level in the expansion tank causing pump cavitation Pump impeller is worn or damaged reducing discharge pressure Air or gas ingress into the suction side of the pump Vapor lock	Reduced flow rate of thermic salt due to lower pump differential pressure Immediate consequence is less heat removal from the reactor leading to hot spots Potential loss of salt inventory from the leak which is a major safety hazard Pump cavitation causing damage to the pump	Install low pressure alarm on the pump discharge Provide a low low pressure interlock that trips the isobutene feed and stops the salt pump Install leak detection systems around the salt piping Use a level transmitter on the expansion tank with low level alarm and	Low pressure alarm Low low pressure interlock for isobutene trip Leak detection system around salt piping Low level alarm and interlock on expansion tank Regular pump maintenance schedule Pump suction pressure monitoring Operator training on leak response procedures

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
			from water or organics entering the hot salt system Control valve is open too far reducing back pressure	impeller and bearings If pressure drops too low the salt may vaporize or degrade Hot salt leak can cause severe burns and fires Operator may not realize there is a leak until level drops significantly	interlock Perform regular pump maintenance and impeller inspection Install a pressure transmitter on the pump suction to detect cavitation conditions Train operators to recognize pressure drop as a sign of leak before level drops	

9.3 Reactor vessel

The temperature profile inside the reactor, mainly temperature at the hotspot, location of the hotspot, catalyst activity, outlet concentrations, pressure drop across the catalyst bed and the residence time.

9.3.1 Catalyst activity

Table 49: HAZOP for the catalyst actvity in the reactor

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
Catalyst activity	More	Catalyst activity is higher than the normal design basis	Fresh catalyst batch has higher than expected activity due to manufacturing variation Higher loading of active phase on the support material Changes in feed composition such as higher	Reaction rate increases significantly at the reactor inlet creating a very sharp temperature peak Hot spot temperature exceeds safe limits leading to catalyst sintering Rapid deactivation of the catalyst due	Characterize each new catalyst batch in a pilot reactor before full loading Design the reactor with a conservative activity allowance such as 20 percent extra cooling capacity Implement a gradual	Pilot scale testing of new catalyst batches Multiple thermocouples along the reactor tubes to detect hot spots Conservative reactor cooling design Operating procedure for catalyst break in period Feed

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
			isobutene concentration promote the reaction Catalyst was regenerated too aggressively increasing surface area Over reduction of the catalyst before startup	to thermal stress Increased risk of runaway exothermic reaction Higher conversion to CO and CO2 instead of desired methacrolein and hence reduced selectivity Shortened catalyst lifetime requiring more frequent replacement	catalyst conditioning or break in procedure during startup Monitor reactor hot spot temperatures closely during the first week of operation Reduce feed temperature or isobutene concentration if activity is too high	isobutene concentration control Ability to reduce reactor inlet temperature
Catalyst	Less	Catalyst activity is	Catalyst has aged beyond	Poor conversion	Monitor reactor	Online gas chromatogra

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
activity		lower than the normal design basis	its useful lifetime due to thermal sintering over many months Poisoning of the catalyst by impurities such as sulfur chlorine or heavy metals in the feed Physical loss of catalyst fines from the tubes due to vibration or pressure cycling Improper catalyst loading leaving voids or channeling Incorrect activation or	of isobutene leading to low yield of methacrolein and MAA High concentration of unreacted isobutene in the reactor outlet which creates an explosion risk downstream Increased byproduct formation due to incomplete oxidation Higher oxygen consumption without correspondin	conversion continuously using online gas chromatography Track catalyst age and historical performance data Implement a catalyst replacement schedule based on predicted lifetime rather than waiting for failure Install impurity monitoring systems on the isobutene and air feed streams	ph for outlet composition Pressure drop indicator across the reactor bed Feed impurity analyzers for sulfur and chlorine Catalyst age tracking system in the maintenance management system Documented catalyst loading procedure with supervision Operator training on

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
			reduction procedure during startup Improper storage of fresh catalyst allowing moisture or contamination to degrade it	g product formation Economic loss due to off spec product and wasted raw materials Need to increase reactor temperature to compensate which accelerates further deactivation Premature catalyst replacement adds cost and downtime	Perform pressure drop measurement across the reactor bed to detect physical degradation of catalyst Train operators on proper catalyst loading procedures	catalyst handling

9.3.2 Temperature profile

Table 50: HAZOP for the temeprature profile in the reactor

Parameter	**Guide word**	**Meaning**	**Causes**	**Consequences**	**Actions required**	**Safe guards**
Temperature deviation from hotspot	More	The axial or radial temperature profile shows a hot spot higher than the normal design maximum temperature	Localized increase in catalyst activity due to fresh catalyst or uneven distribution Uneven flow distribution across the reactor tubes causing some tubes to receive more isobutene than others Partial blockage of thermic salt flow around	Rapid deactivation of catalyst at the hot spot location which then moves deeper into the bed Thermal stress and potential rupture of individual reactor tubes in the hot zone Increased byproduct formation such as CO and CO2 Reduced selectivity	Install multiple thermocouples at different radial positions and axial depths within representative tubes Use a thermal imaging or infrared camera to scan the reactor shell for external hot spots Implement automatic isobutene feed	Multiple thermocouples at different positions in the catalyst bed Temperature difference alarm on DCS Thermal imaging or skin temperature monitoring on the reactor shell Feed flow distributor design at reactor inlet Documented catalyst

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
			certain tubes creating local hot zones Inlet feed temperature is too high pushing the reaction zone upstream Catalyst packing density varies across the bed due to improper loading	to methacrolein and MAA Potential runaway reaction if hot spot propagates through the entire bed Melting or sintering of the catalyst support material Permanent damage to the reactor requiring tube plugging or replacement	reduction or shutdown on high temperature difference between tubes Regularly redistribute feed flow using a flow distributor at the reactor inlet Perform catalyst loading with strict quality control to ensure uniform packing density	loading procedure with vibration to settle bed uniformly Regular operator rounds to compare temperatures across tubes
Temperature deviation	Less	The temperature	Catalyst is deactivated	Poor conversion	Install thermocoupl	Multiple axial

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
from hotspot		profile shows a cold spot or no temperature rise indicating the reaction is not initiating properly	or poisoned near the reactor inlet Feed temperature is too low to activate the reaction Isobutene concentration is too low due to ratio control problem Thermic salt temperature is too high reducing the temperature driving force at the reactor inlet Catalyst was not properly activated or reduced	of isobutene leading to low MAA yield High unreacted isobutene in the reactor outlet creating an explosion risk downstream Inefficient use of oxygen leading to high oxygen concentration in the off gas Potential for thermal runaway if reaction suddenly initiates	es at multiple axial positions to detect where the reaction initiates Increase reactor inlet temperature gradually under controlled conditions until reaction initiates Verify feed composition and ratio control before assuming catalyst is the problem Perform	thermocouples in the catalyst bed Reactor inlet temperature control with ability to increase gradually Feed ratio control system with alarms Catalyst activity monitoring program Operating procedure for startup and catalyst conditioning Off gas oxygen analyzer to

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
			before startup Water or inert gas contamination in the feed dilutes the reactants	further down the bed where cooling is less effective Economic loss due to off spec product and wasted feed	regular catalyst activity testing using a side stream reactor Increase feed temperature setpoint if the reaction zone is consistently moving downstream	detect incomplete conversion

9.3.3 Outlet concentration (conversion to Methacrolein)

Table 51: HAZOP for the outlet concentration in the reactor

Parameter	**Guide word**	**Meaning**	**Causes**	**Consequences**	**Actions required**	**Safe guards**
Concentration of methacrolein in the outlet stream	More	Outlet concentration of methacrolein is higher than expected indicating over conversion beyond the desired intermediate	Reactor temperature is too high pushing the reaction past methacrolein to CO and CO2 Catalyst has high activity for over oxidation such as a different crystal phase or metal impurity Residence time is too long due to low flow rate of feed	Higher oxygen consumption without economic benefit Potential for hot spots in the downstream methacrolein to MAA reactor Catalyst deactivation due to carbon deposition from over oxidation	Install an online gas chromatograph or infrared analyzer for methacrolein CO and CO2 at the reactor outlet Implement an alarm on high CO or CO2 concentration relative to methacrolein Reduce reactor inlet temperature or increase flow rate to	Online gas chromatograph or infrared analyzer for outlet composition Alarm on high CO to methacrolein ratio Reactor temperature control with operator setpoint Flow rate monitoring and control Isobutene to oxygen ratio control system

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
			Oxygen concentration is too high promoting complete oxidation		decrease residence time	Operator training on reaction chemistry
Concentration of methacrolein in the outlet stream	Less	Outlet concentration of methacrolein is lower than expected indicating poor conversion of isobutene or over conversion of methacrolein to the side products	Catalyst activity is low due to aging poisoning or improper activation Reactor temperature is too low to achieve adequate conversion Residence time is too short due to high feed flow rate	Low MAA yield in the downstream reactor because insufficient methacrolein is produced High concentration of unreacted isobutene in the off gas which is flammable and	Install an online analyzer for unreacted isobutene at the reactor outlet Implement an alarm on high isobutene concentration relative to methacrolein Increase reactor inlet temperature gradually	Online gas chromatograph for unreacted isobutene and methacrolein Alarm on high isobutene at outlet Reactor temperature control with ability to increase Feed ratio control

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
			Isobutene concentration is too low due to ratio control problem Feed distribution is uneven across the tubes causing bypassing Thermic salt temperature is too high reducing the temperature driving force	potentially explosive Economic loss due to wasted isobutene and oxygen Increased load on the downstream purification system Potential for isobutene accumulation in the vent system creating an explosion risk Need to increase reactor temperature or replace catalyst	until conversion improves Verify feed ratio and flow distribution before concluding catalyst is the problem Track catalyst age and schedule replacement based on conversion decline rather than fixed calendar time	system with alarms Pressure drop monitoring across reactor bed Catalyst age tracking and performance trending system Operating procedure for conversion troubleshooting

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safe guards
				prematurely		

9.3.4 Reactor pressure

Table 52: HAZOP for the reactor pressure

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safety guards
Pressure inside the reactor	More	Reactor internal pressure is higher than the normal operating range	Blockage at the reactor outlet such as a closed valve plugged line or fouled downstream equipment Catalyst fines have accumulated on the reactor outlet screen	Mechanical stress on the reactor vessel and tube sheet Potential rupture of the reactor shell or tube to tube sheet joints Leakage of flammable and toxic isobutene	Install high pressure alarm on the reactor vessel with automatic action Implement a high high pressure interlock that cuts isobutene feed and opens an	High pressure alarm on DCS High high pressure interlock that trips isobutene feed Pressure relief valve with proper sizing Redundant

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safety guards
			or in the outlet piping Pressure control valve fails closed or is manually closed by operator Feed pressure from upstream compressor is too high	MAA and methacrolein through gaskets or seals Increased pressure drop across the catalyst bed which can fluidize or damage the catalyst Activation of pressure relief devices which vent to flare or scrubber	emergency vent path Install a pressure relief valve sized for fire case and runaway reaction sized for the vessel Provide a pressure transmitter on the reactor outlet to detect blockages early Regularly inspect and clean reactor outlet	pressure transmitters on reactor and outlet line Regular inspection schedule for outlet screens Pressure drop monitoring across catalyst bed

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safety guards
Pressure inside the reactor	Less	Reactor internal pressure is lower than the normal operating range	Leak in the reactor vessel shell head or gaskets Leak in the reactor outlet piping downstream of the reactor Pressure control valve fails open or is manually opened too far Upstream feed compressor or blower fails or trips Suction filter on the feed blower	Loss of flammable and toxic isobutene MAA or methacrolein to the environment which is a major safety hazard Reduced flow rate of reactants leading to poor conversion and low MAA yield Potential for air ingress into the reactor if pressure drops below atmospheric	Install low pressure alarm on the reactor vessel with automatic action Implement a low low pressure interlock that cuts isobutene feed and isolates the reactor from downstream equipment Install flammable and toxic gas detectors around the reactor and outlet piping	Low pressure alarm on DCS Low low pressure interlock for isobutene feed trip Flammable and toxic gas detectors around the reactor Regular leak testing and inspection schedule Operator training on pressure drop investigation Redundant pressure transmitters

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safety guards
			is blocked Low isobutene or air supply pressure from upstream Loss of containment through a relief device that lifted and did not reseat	creating an explosive mixture inside the vessel Difficulty maintaining the correct isobutene to oxygen ratio due to pressure fluctuations Operator may not realize there is a leak until gas detectors activate or level drops in downstream equipment	Perform regular leak testing of reactor flanges gaskets and fittings during maintenance shutdowns Train operators to investigate any unexpected pressure drop immediately Provide a pressure transmitter on the reactor inlet to differentiate between	on reactor inlet and vessel Automatic isolation valve on reactor outlet

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safety guards
					feed and vessel pressure issues	

9.3.5 Space velocity

Table 53: HAZOP for the space velocity in the reactor

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safety guards
Space velocity	More	Space velocity is higher than the normal operating range meaning the gas flow rate per unit volume of catalyst is too high	Feed flow controller fails open increasing isobutene and air flow rate Operator manually increases production rate beyond reactor	Residence time decreases so isobutene has less contact time with catalyst Conversion drops significantly leading to low methacrolei	Install high flow alarm on the total feed flow to the reactor Implement a high high flow interlock that cuts isobutene feed or opens a	High flow alarm on DCS High high flow interlock for isobutene feed trip Maximum flow limit in feed controller Redundant flow

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safety guards
			design capacity Compressor or blower delivers higher than normal flow due to speed controller failure Bypass valve around the reactor is opened allowing some feed to bypass the catalyst bed Multiple feed lines are opened when only one is required Low	n and MAA yield High concentration of unreacted isobutene in the reactor outlet creating an explosion risk downstream Incomplete oxygen consumption leads to high oxygen in off gas which is a safety hazard Catalyst may be fluidized or lifted out of the tubes if	bypass to flare Design the feed control system with a maximum flow limit that operator cannot exceed Install a flow transmitter with redundancy for high integrity trip Regularly monitor pressure drop across the reactor bed as an indirect indicator of space	transmitters Pressure drop monitoring across catalyst bed Operator training on maximum allowable space velocity Compressor speed control with high speed limit

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safety guards
			pressure drop across the reactor indicates channeling or fluidization of catalyst	velocity exceeds settling velocity Increased erosion of catalyst particles generating fines that plug downstream equipment	velocity effects	
Space velocity	Less	Space velocity is lower than the normal operating range meaning the gas flow rate per unit volume of catalyst is	Feed flow controller fails partially closed Compressor or blower delivers lower flow due to suction blockage or	Residence time increases so isobutene spends more time in contact with catalyst Over conversion occurs where	Install low flow alarm on the total feed flow to the reactor Implement a low flow interlock that alerts operator or automatically adjusts	Low flow alarm on DCS Low flow alert with operator response procedure Reactor inlet temperature control with

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safety guards
		too low	mechanical issue Operator reduces production rate without adjusting temperature accordingly Partial blockage in the feed line such as ice or solidified material Filter or strainer on the feed line becomes blocked Low isobutene or air supply pressure from upstream	methacrolein reacts further to CO and CO_2 reducing MAA yield Increased risk of hot spots and runaway reaction because the same heat is released over a smaller catalyst volume Potential for carbon deposition or coking on the catalyst due to prolonged	reactor inlet temperature Reduce reactor inlet temperature proportionally when space velocity decreases to prevent over conversion Monitor outlet CO and CO_2 concentration for signs of over oxidation Maintain minimum flow rate during turndown operations by recycling	ability to reduce during turndown Online CO and CO_2 analyzer at reactor outlet Feed filter and strainer with differential pressure monitoring Bypass or recycle line for turndown operation without reducing flow too much

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safety guards
				contact Higher pressure drop across the reactor because velocity is lower but this is not always linear Temperature profile shifts toward the reactor inlet which may exceed cooling capacity	some product gas	

9.3.6 Pressure drop

Table 54: HAZOP for thepressure drop in the reactor

Parameter	**Guide word**	**Meaning**	**Causes**	**Consequences**	**Actions required**	**Safety guards**
Pressure drop	More	Pressure drop across the catalyst bed is higher than the normal operating range	Catalyst fines or dust have accumulated at the top of the bed or on outlet screens Coke or carbon deposits have built up on the catalyst particles over time Catalyst particles have swelled or broken down due to thermal or chemical	Increased load on the upstream compressor or blower leading to higher energy cost Reduced flow rate of feed through the reactor because compressor cannot overcome the drop This reduced flow then increases residence time leading to over	Install a differential pressure transmitter across the reactor bed with high pressure drop alarm Implement a high pressure drop interlock that alerts operator to reduce feed flow or shut down if critical Perform regular bed pressure drop	Differential pressure transmitter with high alarm High pressure drop alert with operator response procedure Inlet gas filter or strainer with differential pressure monitoring Regular pressure drop trending in operator logs Top screen

Parameter	**Guide word**	**Meaning**	**Causes**	**Consequences**	**Actions required**	**Safety guards**
			degradation Liquid condensation in the reactor such as water wets the catalyst and increases resistance Foreign material such as rust or scale from upstream piping has entered the reactor and blocked the bed Catalyst loading was too tight with excessive packing	oxidation Potential for fluidization or lifting of catalyst if pressure drop becomes high enough to support particle weight Channeling may develop where gas bypasses the high resistance zones leading to poor conversion Mechanical stress on the reactor tube	trending to predict when cleaning or catalyst replacement is needed Install inlet gas filtration to remove particles before they enter the reactor Design the reactor with a top screen that can be cleaned without unloading catalyst Use a pressure drop versus flow correlation	design that allows cleaning access Operating procedure for pressure drop monitoring during normal operation Catalyst replacement schedule based on pressure drop increase

Parameter	**Guide word**	**Meaning**	**Causes**	**Consequences**	**Actions required**	**Safety guards**
			density	sheet and internal supports Risk of reactor tube rupture if pressure drop is localized to a few tubes	to detect anomalies early	
Pressure drop	Less	Pressure drop across the catalyst bed is lower than the normal operating range	Catalyst has been lost from the reactor due to attrition and entrainment out of the tubes A void or channel has developed through the catalyst bed allowing gas	Poor conversion because gas bypasses the catalyst through channels or voids Unreacted isobutene in the reactor outlet creates an explosion risk	Install a differential pressure transmitter with low pressure drop alarm Perform regular catalyst level checks using a level probe or dip tube from the top of	Differential pressure transmitter with low alarm Low pressure drop alert requiring immediate investigation Regular catalyst level measurement procedure

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safety guards
			to bypass large sections of catalyst Catalyst particles have broken down into very fine particles that have been blown out of the reactor Mechanical failure such as a broken tube or damaged tube sheet allows gas to bypass the bed entirely Incorrect catalyst	downstream Low pressure drop may be the first sign of catalyst loss before conversion drops significantly Potential for hot spots in remaining catalyst zones because the same heat is released over less catalyst If a tube has failed there may be cross flow between shell and	the reactor Compare actual pressure drop to expected pressure drop at current flow rate using a performance curve Investigate any unexplained drop in pressure drop immediately including checking for tube leaks using a tracer or thermal imaging	Pressure drop versus flow performance curve for reference Tube leak detection system such as helium tracer or off gas analysis Operator training on pressure drop as a catalyst loss indicator Maximum allowable flow velocity specification in operating manual

Parameter	Guide word	Meaning	Causes	Consequences	Actions required	Safety guards
			loading left gaps or insufficient packing density The catalyst bed has fluidized due to gas velocity being too high	tube sides mixing salt with reactants	Maintain minimum flow velocity to prevent settling but maximum velocity to prevent fluidization	

10. Site and Plant layout

10.1 Site layout

Any project requires planning regarding the positioning and arrangement of the buildings related to the project on the acquired site. The arrangement needs to be systematic and well-thought out as it plays a key role in maintaining safety, good working conditions, efficient transport and easy accessibility to services. A well-planned site layout has both short-term and long-term benefits.

While preparing a Site Layout, various factors and guidelines need to be kept in mind. A few of these are listed as follows, along with some features of the proposed site layout:

4. The main plant should be located at a safe distance from other buildings, calculated based on toxic vapor dispersion modeling (MAA vapors are heavier than air) rather than fire alone. In the event of a runaway polymerization or leak, evacuation must be quick and cause minimum damage. The main plant is positioned centrally but downwind of the control room so that fumes blow away from occupied buildings during an emergency. Multiple access roads from at least three directions are provided.
5. The Control Room and Quality control department is placed near to the main plant so that any deviations in the production process can be identified and suitable measures can be taken quickly. Some land near the main plant can be left for future project expansion purposes.
6. The Tank Farm and raw material storage facilities are situated near the main plant to minimize transfer distances, but with the following mandatory features-
    a. Nitrogen blanketing on all MAA and HEMA storage tanks
    b. Secondary containment dikes sized for 110% of the largest tank
    c. Steam tracing or heated piping since MAA solidifies at 16°C which is a common temperature during Dahej winters.
    d. Vapor return lines connected to the loading/unloading zone
    e. Inhibitor injection points at tank inlets

    f. Loading/unloading zone flooring made of chemically resistant material (e.g., Potassium Silicate concrete)
    g. Tank farm located downwind of the main plant but upwind of the boundary fence
7. There must be provision of emergency facilities like Fire Station, Health Centre, Assembly points, and emergency control room. In case of an emergency, these must be easily accessible. The proposed layout has all these facilities together and near the main plant so that all of these can be accessed during an emergency. There are assembly points provided on two different sides of the plant. Some of the provision of emergency facilities will include -
    a. Fire Station stocked with Alcohol-Resistant Aqueous Film Forming Foam (AR-AFFF) since standard water is ineffective on MAA/HEMA fires.
    b. Health Centre with decontamination showers and atropine (for severe exposure)
    c. Assembly points on two different sides of the plant, both upwind of the main plant under normal wind conditions
    d. Emergency Control Room located outside the toxic vapor dispersion zone
    e. Emergency eyewash stations within 10 seconds (approx. 55 feet / 17 meters) of any MAA/HEMA handling area
    f. All emergency facilities are clustered together and accessible from multiple routes
8. These buildings are located away from the main plant and close to the main gate (Gate 1) , but critically, they are positioned upwind of the plant under prevailing wind conditions to avoid exposure to MAA's pungent, acrid odor. This ensures minimum interference in plant operations and prevents unauthorized personnel from entering hazardous zones. R&D labs working with MAA/HEMA have a separate, smaller dedicated entrance.
9. Parking is kept close to Gate 1 and beside the main administrative building. It is situated outside the hazardous zone and away from the main plant to ensure that smaller vehicles (which can be ignition sources) do not interfere with plant operations. No parking is allowed near the tank farm or loading/unloading zone.
10. Roads are minimum 9 meters wide to allow unrestricted movement of tank trucks, fire tenders, and mobile cranes. The road network is designed with-

    a. Minimum number of turns to reach the tank farm, loading/unloading zone, and main plant
    b. No sharp bends near heated MAA transfer lines
    c. Separate one-way loops for truck entry (Gate 2) and exit (Gate 2 or Gate 3)
    d. Emergency vehicle lanes with a minimum clear height of 4.5 meters
    e. No speed bumps in truck corridors to prevent surge damage to heated hoses
11. There are different designated gates in the proposed layout based on the purpose they serve. Gate 1 is for the daily employees and visitors and hence, it is closest to the administrative building. Gate 2 serves the purpose of materials being transported to and from the plant which would usually see the movement of trucks near the loading/unloading zones. Gate 3 can act as an emergency exit and can be used if there is a situation in which Gate 1 or 2 cannot be used. All security cabins are equipped with gas detectors for MAA vapors and communication links to the Emergency Control Room.
12. Green belts are provided around the site where trees and plants can be planted. But there are a few points to be considered with respect to the plant species
    a. Non-flammable (avoid trees with high resin/sap content near electrical areas)
    b. Dense foliage to act as a passive vapor dispersion barrier
    c. Evergreen for year-round effectiveness
    d. Minimum width of green belt between plant and boundary: 15 meter
13. There is a provision of an emergency water supply which may be required in certain circumstances.
14. The utilities and the electric power station have also been allocated some spaces in the layout, along with the Effluent Treatment Plant. The utilities required will be -
    a. Electical power station
    b. Cooling tower
    c. Steam boiler
    d. Nitrogen generation and storage
    e. ETP
    f. Hazardous waste storage

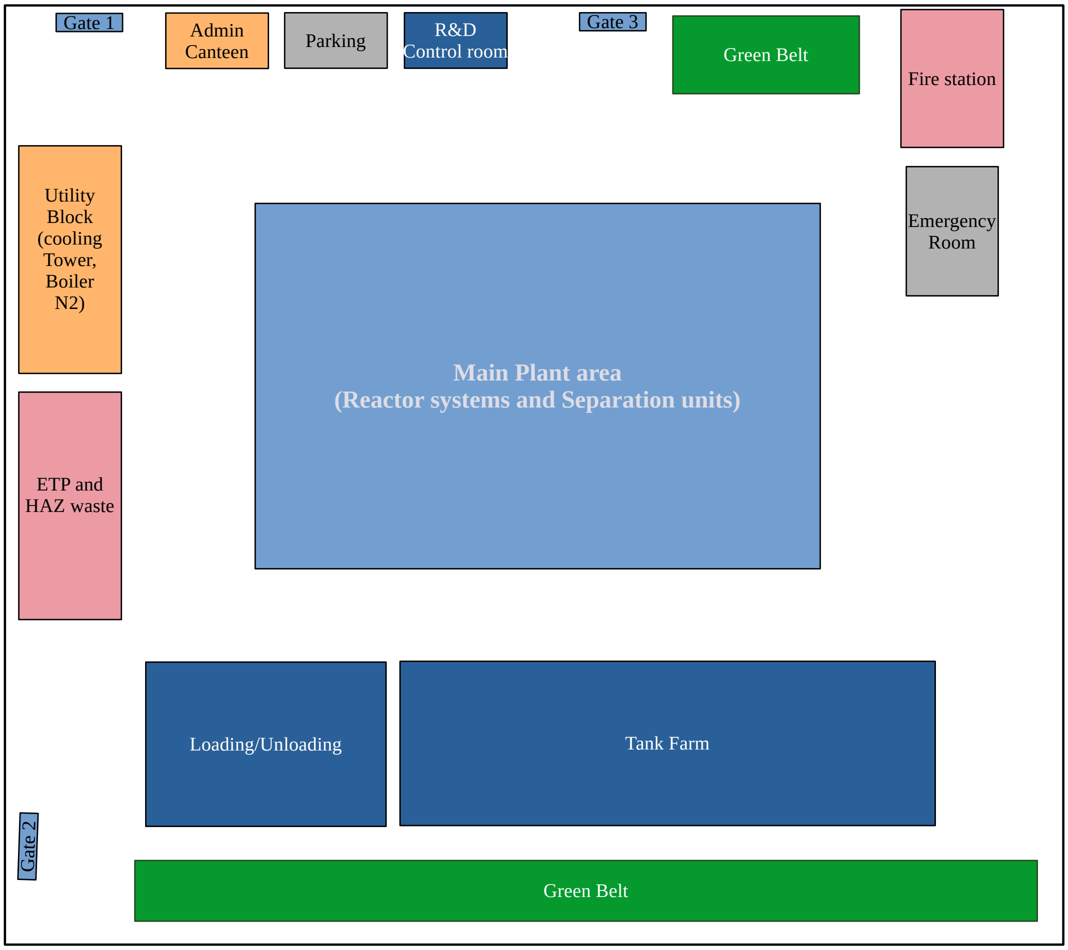


Figure 7: Site layout

10.2 Plant Layout

The Plant layout involves the systematic arrangement of equipment involved in the production process and located in the main plant of the project site.

Some guidelines to be followed while making a plant layout are:

4. The arrangement should not impede the flow and sequence of the production process.

5. There must be sufficient space allocated for the equipment along with space enabling easy access for the operators, considering the safety factor.
6. The costs can be minimized by placing the equipment such that the piping length is optimized and use gravity flow wherever possible to reduce pumping costs. Tanks can be placed at heights to allow for gravity flow.
7. Arrangement should allow easy maintenance. If any equipment requires dismantling, then sufficient space near it must be present.
8. A localized plant office must be present to monitor the daily operation of the plant.
9. Safety provisions like safety showers, a medical clinic, and escape routes must be present, which can be utilized in the event of an accident.
10. The plant should be ready for future expansion without significant changes.
11. The plant is usually divided into sections/areas based on the processes being carried out.

11. Safety, Health and Environement

While undertaking a chemical project, whether it involves manufacturing, research and development, or any other aspect of the chemical industry, integrating safety, health, and environmental considerations are crucial. This leads to minimization of risks to personnel, protection of environment and ensures sustainable practices within industry. Government and regulatory bodies often impose rules and guidelines which need to be followed during the lifetime of a chemical project.

The NFPA Diamond under NFPA 704 is a standard system for the identification of the hazards of the materials used. It used to facilitate an appropriate response to an emergency. It is a color-coded number system in which the hazards associated with a chemical are assigned numbers based on the degree of health hazard (blue), flammability hazard (red) and reactivity/instability hazard (yellow). These are represented in the form of a diamond with 4 quadrants containing the color-codes as shown below

Figure 8: NFPA diamond

Table 55: NFPA ratings for chemicals used

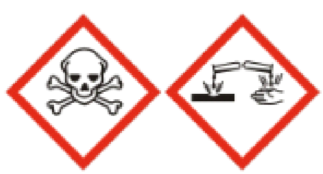
Chemical	Health	Flammability	Stability
Isobutene	3	4	1
Ethylene oxide	4	4	3

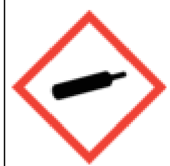
Chemical	Health	Flammability	Stability
N-butyl acetate	2	3	0
Methacrolein	3	3	2
Methacrylic acid	2	2	1
HEMA	3	2	1

11.1 Safety

Human safety shall override all production targets. Hazards must be promptly identified and appropriately mitigated to prevent accidents. Beyond economic losses, accidents can severely impact an organization's reputation. Therefore, suitable safety measures must be installed throughout the plant to ensure a consistently high level of safety. Chemical hazards are commonly communicated pictorially through Globally Harmonized System (GHS) labels. Each GHS label features a red diamond containing a black-and-white pictorial representation of the specific hazard. By referring to these labels, personnel can quickly identify the potential risks associated with any chemical used in the facility. The GHS labels for all chemicals utilized in this process are shown below.

Table 56: GHS labels for the chemicals used

Chemical	GHS label	Signal word
Isobutene		Danger
Ethylene oxide		Danger
N-butyl acetate		Warning
Methacrolein		Danger
Methacrylic acid		Danger
HEMA		Warning

Oxygen		Danger
Nitrogen		Danger

Common safety measures that need to be present are -

1. Fire-fighting equipment shall be present in adequate quantities at appropriate locations. Due to the polar nature of MAA and HEMA, Alcohol-Resistant Aqueous Film Forming Foam (AR-AFFF) extinguishers and systems are mandatory, as standard CO_2 or dry chemical powder extinguishers alone are insufficient for polar solvent fires. Carbon dioxide (CO_2) and dry chemical powder type extinguishers shall be provided for electrical fires. Fire alarms and smoke detectors must be installed at multiple locations, especially near the reactor, distillation columns, and tank farm, and maintained in working condition at all times. Water sprinkler systems shall be installed for cooling of adjacent equipment, but water alone shall not be used on large MAA/HEMA pool fires due to the risk of boiling and spreading the burning liquid.

2. Emergency call points, sirens, and communication systems must be readily accessible to all working personnel. Given the toxicity of MAA (GHS H311: Toxic if swallowed) and its pungent, low-odor threshold, emergency communication devices shall be located at every exit from the main plant area, the tank farm, and the loading/unloading zone. Multiple sirens shall be placed such that they can be heard above normal plant operating noise levels.

3. Electrical equipment shall be insulated with appropriate insulating materials to prevent electrical shocks. All electrical equipment in areas where MAA, HEMA, or their vapors may be present shall be rated for Class I, Division 2 (or higher, based on the flammability ratings of intermediates like isobutene and methacrolein). Electrical equipment shall be kept isolated from flammable materials, as even a small spark can cause fire or explosion. Earthing shall be provided for all electrical and metallic equipment to prevent static charge accumulation, which is particularly critical during tank truck loading of MAA and HEMA. High-voltage devices and wiring must be located away from personnel access areas and routed outside hazardous zones wherever possible.

4. Machines and equipment within the plant present associated mechanical hazards. These shall be spaced appropriately with ample walking space around them, with a minimum clearance of 1 meter around rotating equipment. Moving parts such as motors, agitator shafts, rotor blades, and gears must be covered with fixed guards to prevent human contact during operation. Abnormal process conditions, including high temperatures, can induce exothermic polymerization in both MAA and HEMA, leading to rapid pressure buildup and potential rupture of equipment. Therefore, regular checking of the Material of Construction (MOC) for corrosion or stress cracking must be performed, as MAA is corrosive to carbon steel and requires stainless steel (typically 316L) for safe service.

5. Pipelines and equipment shall be regularly inspected for any leakages, with special attention to flange joints, pump seals, and valve packing in MAA and HEMA service. If leakages are detected, they must be addressed without delay using appropriate personal protective equipment. Material Safety Data Sheets (MSDS) for all chemicals shall be present within the plant at every location where the chemical is used, including the main plant, tank farm, loading/unloading zone, and control room. Proper ventilation must be provided to prevent accumulation of hazardous vapors, as MAA vapors are heavier than air and can accumulate in low-lying areas. Continuous gas detection shall be installed for methacrolein and isobutene in the reactor area. Oxygen levels shall be maintained between 19.5% and 23.5% for operator safety during normal operations and maintenance activities.

6. Correct labels, signs, symbols, and warnings must be clearly displayed wherever hazards are present. GHS-compliant labels showing health hazard (H311, H314), corrosion, and environmental toxicity shall be posted on all MAA and HEMA storage tanks, piping, and at entry points to the plant area. Specific warnings regarding the risk of uncontrolled polymerization shall be displayed near the reactor and monomer storage tanks.

7. Emergency evacuation routes shall be planned during the construction phase of the plant. For MAA and HEMA, which produce vapors heavier than air, evacuation routes shall be oriented perpendicular to the prevailing wind direction and shall lead to assembly points located upwind of the plant under normal conditions. At least two distinct evacuation paths shall be provided from each work area. Evacuation routes shall be clearly marked with illuminated signs and maintained free of obstructions at all times.

8. All employees and personnel must be made aware of the specific hazards of MAA and HEMA, including their corrosive nature, toxicity, and potential for runaway polymerization. They shall receive sufficient training to take appropriate action in the event of an emergency, including the correct use of AR-AFFF fire extinguishers, operation of emergency showers and eyewash stations (located within 10 seconds or approximately 55 feet of any MAA/HEMA handling area), and execution of emergency shutdown procedures for the reactor cooling system to prevent polymerization.

11.2: Health

The chemicals used in the plant can pose various health hazards and can have adverse effects on the working personnel if they are exposed to the chemicals. Material Safety Data Sheets list out the health hazards of the chemicals and often state appropriate measures to be taken upon exposure. The MSDS of all the chemicals used in the plant must be accessible and the personnel must be trained to understand these sheets.

Personal Protective equipment (PPE) must be used whenever required. Some of these are:

4. Face shields, masks, and helmets

5. Safety eye goggles and ear plugs
6. Safety uniform, IFR suits, gloves
7. Safety shoes or boots
8. Respirators and self-contained breathing apparatus

Safety showers and eye baths must be present at accessible locations inside the plant. First-aid kits must be present at regular intervals in the plant. A health center/clinic must be present inside the plant with trained people to treat personnel and monitor their health.

The phone numbers of nearby hospitals and emergency fire stations must be listed out and the emergency facilities must be readily available in case of any exigent circumstances.

11.3: Environment

The chemical industry is shifting to sustainable practices, and environmental protection is becoming a priority. Regulatory authorities impose rules and guidelines which need to be followed to minimize environmental impact. The usage of materials should be reduced whenever possible. Water needs to be recycled and proper rainwater harvesting systems should be installed. Use of renewable sources of energy in the form of solar panels should be done. The site should have multiple green belts where trees and plants can be planted.

Water quality and groundwater availability need to be checked regularly to prevent contamination. Land and soil pollution must be avoided while constructing the plant and during operations. Emissions from the plant also need to be monitored and minimized.

11.3.1 Effluent treatment

The effluent treatment will be required for the raffinate of the extraction column T-303. This stream is a high-strength organic wastewater with two major challenges - high COD from n-Butyl Acetate and toxicity/poor biodegradability of Methacrylic acid (MAA).

Given the complexity of the mixture, here is the specific behavior of each chemical in the proposed treatment system -

- n-Butyl Acetate (1,433.3 mg/L - The largest contaminant): This is highly treatable. It is a Volatile Organic Compound (VOC) that is readily removed via gas-liquid absorption and subsequent biological degradation . The anaerobic digester will easily break this down into methane and CO2.

- Methacrylic Acid (57.3 mg/L - The primary product): This is the bottleneck. It is toxic to microorganisms, but industry solutions exist. Research indicates that polymerized MAA membranes are used for water remediation, but the monomer itself inhibits biomass . Therefore, a chemical detoxification pre-treatment is essential to break its toxic structure before biological treatment .

- Acrylic Acid (0.1 mg/L - Trace): Although present in very low amounts, it is highly toxic to methanogens (anaerobic bacteria) at higher concentrations . However, because it is a trace, it will likely be successfully degraded during anaerobic treatment .

- Water (2,714.8 mg/L - The carrier): The actual volume of water is low relative to the organics, confirming this is a high-strength industrial effluent.

So the effluent treatment will have to consider the following points -

1. Detoxification is Mandatory (The MAA Problem): Do not send this directly to a biological plant. The MAA will likely "shock" the bacteria. You must install a pre-treatment reactor to polymerize or neutralize the MAA, or dilute the stream significantly with other plant wastewater .

2. pH Control is Crucial: MAA and Acrylic Acid are corrosive. Your equalization tank must have chemical-resistant lining and automated dosing pumps to neutralize the acidity before it destroys the concrete or the bacterial biomass.

3. Anaerobic Granulation Technology: For the high-strength n-Butyl Acetate, using granular sludge (UASB reactor) is recommended. The dense granules are resistant to the shock loads and toxicity of this type of chemical waste .

4. Potential for Odor/VOC: The n-Butyl Acetate has a strong fruity odor and is volatile. The equalization tanks and anaerobic reactors should be covered and the off-gases treated (e.g., by a biofilter or caustic scrubber) to prevent nuisance odors.

The block diagram for the ETP is given in

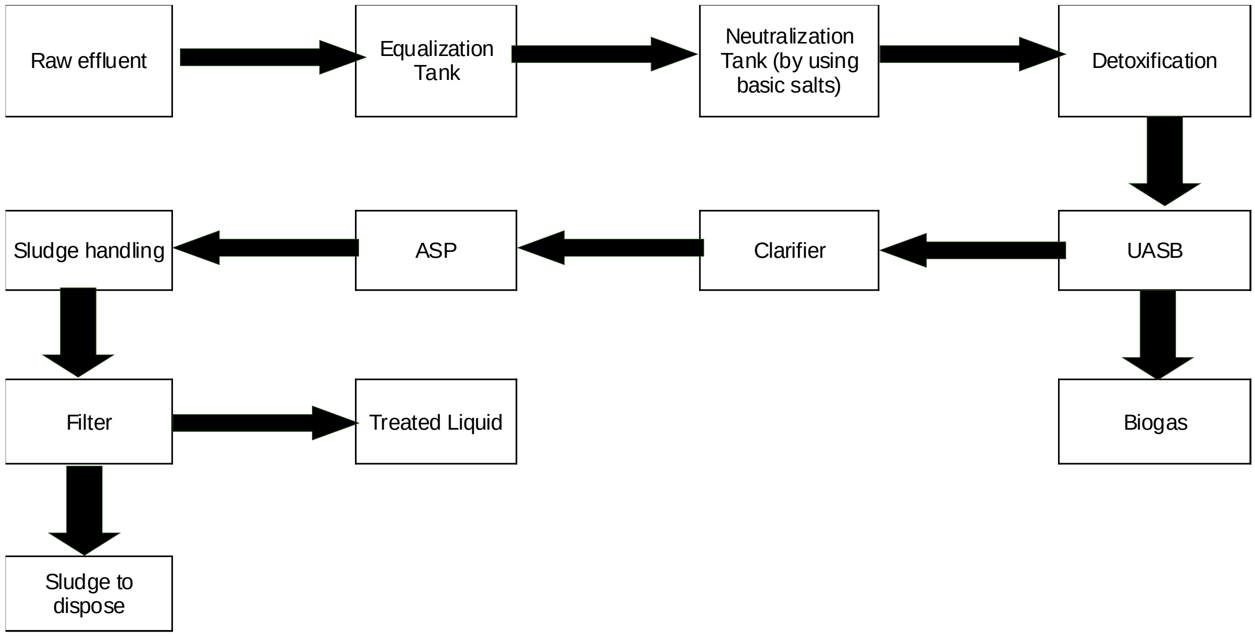


Figure 9: ETP block diagram

11.3.2 Flare system

The gas outlet from the condenser C-301 will have to be sent to a flare as it contains air and a large amount of organics. Before sending to the flare, the air can be adsorbed using PSA technique and then can be reused and recylced.

11.3.2 Solid waste (TPA)

The terepthallic acid formed in reactor R-202 will have to be filtered out before the stream is sent for further purification to avoid fouling and further degradation. The amount of TPA generated is 6 kg/h, which can be stored and sold at a bare minimum cost if possible, since the purity of the TPA will be almost 100%.

12 Project cost

Project costing involves the estimation of initial investment required to set up the project. The cost of all the equipment and machinery needs to be found first and then the total project cost is determined based on the plant and machinery costs.

12.1 Cost of Material of construction

The current costs of the MOC were found out in available literature and purchaser databases and are given below in Table 57: Cost of Material of Construction

Table 57: Cost of Material of Construction

Material	**Cost (Rs/kg)**
SS316L	400
Hastealloy	4000
Carbon steel with PTFE lining	5000 Rs per m^3

12.2 Cost of equipment

Since we had all the required data for the costing, the equipment cost was done by calculating the weight of the equipment multiplied the cost. This gave an exact cost for all the equipments.

12.2.1 Costing of Isobutene reactor system

Table 58: Isobutene Reactor system costing summary

	R201	**R202**	**C301**	**C203**

Type	Multitubular packed bed reactor	Multitubular packed bed reactor	Condenser	Crystallizer
Volume (m3)	4.87	4.87	10	6.5
Material	SS316	SS316	SS316	SS316
Density (kg/m3)	7,980	7,980	7,980	7,980
Weight (kg)	38,863	38,863	79,800	51,870
Cost per kg (Rs)	400	400	400	400
Fabrication cost per kg (Rs)	133	133	133	133
Total cost per kg (Rs)	533	533	533	533
Total cost (Rs)	**20,726,720**	**20,726,720**	**42,560,000**	**27,664,000**

12.2.2 Costing of Separation units for MAA

Table 59: Separation units costing summary

Type	T204	T302	T303	T304

	LLE column	Distillation column	Distillation column	Distillation column
Volume (m3)	0.043	0.234	0.265	0.136
Material	SS316	SS316	SS316	SS316
Density (kg/m3)	7,980	7,980	7,980	7,980
Weight (kg)	343	1,867	2,115	1,085
Cost per kg (Rs)	400	400	400	400
Fabrication cost per kg (Rs)	133	133	133	133
Total cost per kg (Rs)	533	533	533	533
Total cost (Rs)	**183,008**	**995,904**	**1,127,840**	**578,816**

12.2.3 Cost for equipments for HEMA production

The impeller and the helical coil for cooling the reactor is made of Hastealloy since ethylene oxide is a reactant and there needs to be utmost safety precaution. To reduce the cost, a PTFE lining Carbon steel is used for the shell vessel.

Table 60: Reactor system for HEMA production costing summary

	R203	**R204**	**R205**	**R206**
Type	**CSTR gas liq**	**CSTR gas liq**	**CSTR gas liq**	**CSTR gas liq**
Volume (m3)	1.159	1.159	1.159	1.159
Material	CS +PTFE lining	CS +PTFE lining	CS +PTFE lining	CS +PTFE lining
Total cost per m3	5000	5000	5000	5000
Total cost	**5795**	**5795**	**5795**	**5795**

Table 61: Cost for the imepller and the helical coil

Type	**R203**	**R204**	**R205**	**R206**

	CSTR gas liq	CSTR gas liq	CSTR gas liq	CSTR gas liq
Volume (m3)	0.05	0.05	0.05	0.05
Material	hastealloy	hastealloy	hastealloy	hastealloy
Density (kg/m3)	8,890	8,890	8,890	8,890
Weight (kg)	445	445	445	445
Cost per kg (Rs)	4,000	4,000	4,000	4,000
Fabrication cost per kg (Rs)	1,333	1,333	1,333	1,333
Total cost per kg (Rs)	5,333	5,333	5,333	5,333
Total cost (Rs)	**2,370,667**	**2,370,667**	**2,370,667**	**2,370,667**

12.2.4 Catalyst Cost

The metal oxide mixed catalyst used for MAA production and the $FeCl_3$ catalyst for HEMA production can be included in the equipment costs since the have the life cycle of 10 years (minimum). Even the thermic salt used as a heat trasnfer fluid is included in these costs, since it will recycled continuously without any waste as it is very expensive and it would be uneconomical to not do so.

Table 62: Catalyst and heat transfer fluid cost

Catalyst	**Mass (kg)**	**Cost per kg**	**cost for 5 years**
R201	21618.89	15	₹ 324.28 L
R202	21618.89	15	₹ 324.28 L
R203-6	1319.37	30	₹ 39,581.10
Thermic salt	560000	400	₹ 224.00 Cr
		Total	**₹ 224.69 Cr**

Thus, adding all the costs, we get a fixed cost for the equipments as ₹ **349.61 Cr.**

12.3 Cost of Plant and machinery

Based on the estimated gross equipment costs, the costs related to the plant and machinery such as Piping, Instrumentation, Electrical costs etc are calculated. These costs are estimated as a percentage of the gross equipment cost. The percentages used are as per the guidelines in the cost estimation of chemical projects. The table below shows the costs of plant and machinery.

Table 63: Cost of Plant and machinery

Cost Of Plant And Machinery

Cost Component	% Of Equipment Cost	Cost (Rs)
Total Equipment Cost	-	₹ 349.61 Cr Rs
Piping	30	₹ 104.88 Cr Rs
Instrumentation	20	₹ 69.92 Cr Rs
Electrical	20	₹ 69.92 Cr Rs
OSBL Facilities	30	₹ 104.88 Cr Rs
Sub -Total A	A	₹ 349.61 Cr Rs

Spares	5% Of A	₹ 17.48 Cr Rs
Sub - Total B	B	₹ 367.09 Cr Rs
Cost Components	% Of B	Cost (Rs) Rs
Packing And Forwarding	3	₹ 11.01 Cr Rs
Transportation	3	₹ 11.01 Cr Rs
Installation	20	₹ 73.42 Cr Rs
Painting And Insulation	5	₹ 18.35 Cr Rs
Insurance	1	₹ 3,670.89 L Rs
GST	18	₹ 66.08 Cr Rs
Project Import Duty	5	₹ 18.35 Cr Rs
Grand Total		₹ 918.60 Cr Rs

12.4 Project cost

For this plant, since it is a part of an already existing pterochemical complex, the cost of the plant machinery was assumed to be a higher percentage of the total cost and then the total project cost was calculated as in Table 64: Total Project Cost.

Table 64: Total Project Cost

Total Project Cost		
Cost Component	**% Of Total Cost**	**Cost (Rs)**
Plant And Machinery	50	₹ 918.60 Cr Rs
Land And Site Development	2	₹ 36.74 Cr Rs
Building And Civil Work	10	₹ 183.72 Cr Rs
Know-How And Engineering Costs	10	₹ 183.72 Cr Rs
Miscellaneous Fixed Assets	3	₹ 55.12 Cr Rs
Pre-operative Expenses	10	₹ 183.72 Cr Rs
Contingencies	10	₹ 183.72 Cr Rs

Preliminary And Capital Issue Expenses	2	₹ 36.74 Cr Rs
Margin Money	3	₹ 55.12 Cr Rs
		Rs
Total Cost	100	₹ 1,837.19 Cr Rs
		Rs
Thus, The Total Project Cost Is Estimated To Be		₹ 1,837.19 Cr Rs

13. Cost of Production

This plant is expected to produce 23,000 TPA of HEMA and is operated continuously for 8000h in a year. Each day a total of 72 tonnes of HEMA is produced. For producing the plant as per capacity, there will be various costs involved. stimation of the yearly operating costs plays a crucial role in determining the selling price of the product such that it is profitable. The cost of production involves Fixed costs in the form of Labor required and factory overhead costs which are incurred irrespective of the amount of product manufactured. Variable costs consist of the raw materials, utilities and packing of the product which depend on the amount of production. Other costs in the form of administration and sales costs are also incurred on a yearly basis.

13.1 Raw material cost

The raw materials involved in the production will only be the compressed air and the ethylene oxide. The cost of isobutene is assumed to be half of the market cost since it is obtained from an already existing cracker in the petrochemical complex. Based on the daily requirement, the yearly cost of raw materials was calculated.

Table 65: Raw material cost

Raw Materials	**No. Of Days for storage**	**Amount Required (Kg/day)**	**Cost per kg**	**Cost/day (Rs)**
Isobutene	3	56,112.00	40	₹ 2,244.48 L
Air	1	735,652.36	0.8	₹ 588.52 L
Ethylene oxide	20	24,186.11	86	₹ 2,080.01 L
	Total Raw Material Inventory Costs =			**₹ 4,913.01 L**

Maintenance And Spares Cost	1%	₹ 49,130.07
Other Fixed Costs	10%	₹ 491.30 L
Total Working Capital per day		**₹ 4,918.41 L**
Total Working Capital per year		**₹ 1,639.47 Cr**

13.2 Utilties Cost

Based on the requirement of the utilities such as cooling water and steam, the cost was estimated.

Table 66: Utility costing

Utility	Requirement (kg/h)	Cost (Rs/kg)	Cost per year
Cooling water	26,791.86	1.00	₹ 214.33 Cr
Steam	2,124.53	0.80	₹ 13.60 Cr
Polymerization inhibitor	60.75	100.00	₹ 48.60 Cr

Total Cost	**₹ 276.53 Cr**

13.3 Gross cost of production

The variable costs roughly account for 60-80% of the gross cost of production. Considering that the total variable costs (due to raw materials, utilities and packaging) account for 70% of the total cost of production, the fixed and other costs can be estimated based on their percentage contribution.

Table 67: Gross cost of production

Type	**Component**	**Cost (Rs)**	**% Contribution**
		Per year	
Variable Costs	Raw Materials	**₹ 1,639.47 Cr**	64.18
	Utilities	**₹ 276.53 Cr**	10.82
	Total Variable Costs	**₹ 1,916.00 Cr**	**75**

Fixed Cost	Labour	**₹ 255.47 Cr**	10
	Factory Overheads	**₹ 255.47 Cr**	10
	Total Fixed Costs	**₹ 510.93 Cr**	
Others	Admin And Sales	**₹ 127.73 Cr**	5
	Total	**₹ 2,554.67 Cr**	**100%**

So, **₹ 2,554.67 Cr** will be the gross production cost to produce 23,000 TPA of HEMA.

14. Financial analysis

Financial analysis is carried out to check the flow and distribution of money with respect to time and to check whether the project is profitable or not. The components of financial analysis are summarized in Table 68: Components of Project financing

Table 68: Components of Project financing

Component	Value
Total Fixed Capital Investment	₹ 1,837.19 Cr
Debt To Equity Ratio	1
Term Loan	₹ 918.60 Cr
Equity Participation	₹ 918.60 Cr
Borrowed Working Capital	₹ 367.44 Cr

14.1 Term Loan repayment

Term loan of **₹ 918.60 Cr** is to be repaid in 5 yearly installments. The interest rate charged is 12% per annum and the interest is calculated using simple interest. Based on this, the amount paid and the principal remaining after each installment is shown in Table 69: Term loan Repayment.

Table 69: Term loan Repayment

Instalment	Principle	Interest	Loan	Principle

		Payable	Repayment With Interest	Left
0	₹ 918.60 Cr	-	-	-
1	₹ 918.60 Cr	₹ 110.23 Cr	₹ 293.95 Cr	₹ 734.88 Cr
2	₹ 734.88 Cr	₹ 88.19 Cr	₹ 271.90 Cr	₹ 551.16 Cr
3	₹ 551.16 Cr	₹ 66.14 Cr	₹ 249.86 Cr	₹ 367.44 Cr
4	₹ 367.44 Cr	₹ 44.09 Cr	₹ 227.81 Cr	₹ 183.72 Cr
5	₹ 183.72 Cr	₹ 22.05 Cr	₹ 205.77 Cr	₹ 0.00

14.2 Borrowed Working Capital

The borrowed working capital of **₹ 367.44 Cr** is taken as a short-term loan of 5 years. The simple interest is calculated by considering an annual interest rate of 15%. The 5 yearly installments are paid as shown in Table 70: Borrowed working capital repayment

Table 70: Borrowed working capital repayment

Instalment	Principle	Interest Payable	Loan Repayment With Interest	Principle Left

0	₹ 367.44 Cr -	-	-	
1	₹ 367.44 Cr	₹ 55.12 Cr	₹ 128.60 Cr	₹ 293.95 Cr
2	₹ 293.95 Cr	₹ 44.09 Cr	₹ 117.58 Cr	₹ 220.46 Cr
3	₹ 220.46 Cr	₹ 33.07 Cr	₹ 106.56 Cr	₹ 146.98 Cr
4	₹ 146.98 Cr	₹ 22.05 Cr	₹ 95.53 Cr	₹ 73.49 Cr
5	₹ 73.49 Cr	₹ 11.02 Cr	₹ 84.51 Cr	₹ 0.00

14.3 Depreciation Schedule

Depreciation involves dividing the assets into project cost components like Land, Buildings, Plant and machinery and other fixed assets. The land value does not depreciate and is not included in the depreciation schedule. The depreciation rates for these components vary. Plant and machinery generally have higher depreciation rates than other assets as they are exposed to harsh conditions and undergo wear and tear during operation.

Depreciation is usually calculated using two methods, namely, Straight Line Method (SLM) and Written Down Value (WDV) method.

In case of straight-line method, the asset depreciates at a constant rate and the value after n years is given by:

$$V_a = V_0 – (D_{SLM} * V_0 * n)$$

Whereas, when using Written Down Value method, the asset value after n years is given by:

$$V_a = V_o * (1 - D_{WDV})^n$$

Where V_0 = Original Value of asset, D_{SLM} and D_{WDV} are the depreciation rates

The depreciation of the cost components of the project is calculated by using the following depreciation rates in Table 71: Depreciation rates.

Table 71: Depreciation rates

Cost Components	Value	DSLM (%)	DWDV (%)
Land And Site	₹ 36.74 Cr Rs	0	0
Building And Civil Works	₹ 183.72 Cr Rs	3.34	10
Plant And Machinery	₹ 918.60 Cr Rs	10.34	25
Miscellaneous Fixed Assets	₹ 55.12 Cr Rs	3.34	10

The values of depreciation and the assets calculated using the 2 methods for 10 years, assuming no salvage cost, are shown in the tables given below.

Table 72: Asset value using SLM

Year	Building And Civil Works	Plant And Machinery	Miscellaneous Fixed Assets	Total
0	₹ 177.58 Cr	₹ 823.61 Cr	₹ 53.27 Cr	₹ 1,054.47 Cr
1	₹ 171.45 Cr	₹ 728.63 Cr	₹ 51.43 Cr	₹ 951.51 Cr
2	₹ 165.31 Cr	₹ 633.65 Cr	₹ 49.59 Cr	₹ 848.55 Cr
3	₹ 159.17 Cr	₹ 538.66 Cr	₹ 47.75 Cr	₹ 745.59 Cr
4	₹ 153.04 Cr	₹ 443.68 Cr	₹ 45.91 Cr	₹ 642.63 Cr
5	₹ 146.90 Cr	₹ 348.70 Cr	₹ 44.07 Cr	₹ 539.67 Cr
6	₹ 140.77 Cr	₹ 253.72 Cr	₹ 42.23 Cr	₹ 436.71 Cr
7	₹ 134.63 Cr	₹ 158.73 Cr	₹ 40.39 Cr	₹ 333.75 Cr
8	₹ 128.49 Cr	₹ 63.75 Cr	₹ 38.55 Cr	₹ 230.79 Cr
9	₹ 122.36 Cr	₹ -31,232,248.77	₹ 36.71 Cr	₹ 127.83 Cr

Table 73: Asset value using WDV

Year	Building And Civil Works	Plant And Machinery	Miscellaneous Fixed Assets	Total
0	₹ 165.35 Cr	₹ 688.95 Cr	₹ 49.60 Cr	₹ 903.90 Cr
1	₹ 148.81 Cr	₹ 516.71 Cr	₹ 44.64 Cr	₹ 710.17 Cr
2	₹ 133.93 Cr	₹ 387.53 Cr	₹ 40.18 Cr	₹ 561.64 Cr
3	₹ 120.54 Cr	₹ 290.65 Cr	₹ 36.16 Cr	₹ 447.35 Cr
4	₹ 108.48 Cr	₹ 217.99 Cr	₹ 32.55 Cr	₹ 359.02 Cr
5	₹ 97.64 Cr	₹ 163.49 Cr	₹ 29.29 Cr	₹ 290.42 Cr
6	₹ 87.87 Cr	₹ 122.62 Cr	₹ 26.36 Cr	₹ 236.85 Cr
7	₹ 79.09 Cr	₹ 91.96 Cr	₹ 23.73 Cr	₹ 194.77 Cr
8	₹ 71.18 Cr	₹ 68.97 Cr	₹ 21.35 Cr	₹ 161.50 Cr
9	₹ 64.06 Cr	₹ 51.73 Cr	₹ 19.22 Cr	₹ 135.01 Cr

14.4 Sales realization

The annual production is 23000 metric tonnes. This is sold at the market at 200 Rs/kg. Therefor the total yearly sales = **₹ 4,763.93 Cr.**

14.5 Break-even Analysis

The break-even capacity of the plant is the capacity at which there is no-profit and no-loss considering the sales and incurred costs of production. To be profitable, the plant needs to be operated at a capacity that is greater than the estimated break-even capacity.

While determining the break-even capacity, the following assumptions are made:

• All expenses are bifurcated into fixed and variable costs.

• The market conditions are ideal. Whatever is produced is sold immediately.

• The selling expenses are fixed as some percent of sales.

• The following annualized (fixed) costs are not dependent on capacity utilization: Interest, charges on term loan, depreciation, salary and wages of plant personnel, factory overheads and administrative expenses.

The break-even capacity is calculated using the following equation:

$$S * X = V * X + F$$

Where, X = break-even capacity, S = price per ton of product, V = variable cost of production per ton of product, and F = Fixed cost of production.

shows the estimation of break-even capacity.

Table 74: Break-even analysis

Fixed Cost Of Production (F)	₹ 918.60 Cr Rs
Variable Cost Of Production (V)	₹ 80,437.88 Rs/ Ton
Price Of The Product Per Ton (S)	200000 Rs/Ton
Breakeven Capacity (X)	7,683.00 TPA
	32.25 % Of The Installed Capacity

14.6 Ratio analysis

Investment decision and profitability for a manufacturing project is often checked using a variety of ratios and indicators. These ratios are mainly divided into Performance ratios and Financial Ratios.

14.6.1 Performance Ratios

Payback Period:

The payback period is determined for 100% capacity utilization.

Gross Profit per year = (Sales Revenue) - (Annual cost of production) = **₹ 2,209.26 Cr**

Total capital investment = Total project cost + Borrowed working capital = **₹ 1,286.03 Cr**

Return on Investment (ROI):

ROI = (Gross profit per year / Total capital investment) * 100 = **171.8 %**

Profit margin = (Gross annual profit / Annual sales revenue) = **46.4 %**

Net assets turnover = Sales revenue / Total capital investment = **2.59**

14.6.2: Financial Ratios

Debt to Equity Ratio = 1:1

Interest cover = Gross profit / Interest

Interest paid after first year = interest on term loan + interest on borrowed capital = ₹ 28.81 Cr

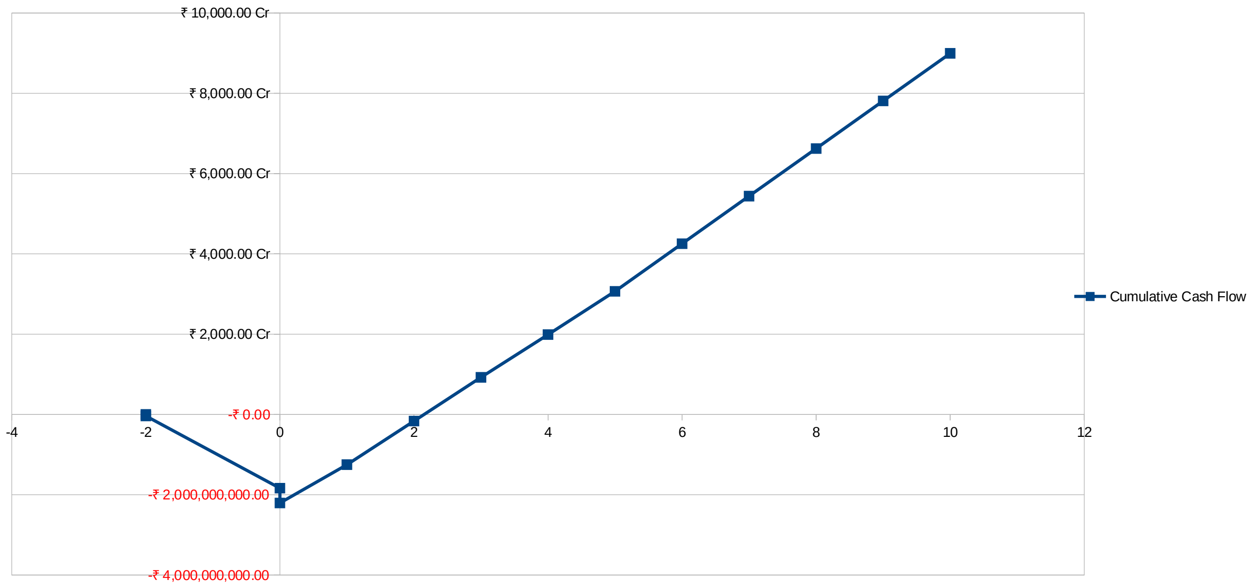
14.7: Estimates of working results

The projected working results for 10 years of plant operation are summarized in annexure 4. The various terms expressed in the table are explained as follows:

3. Gross Profit = Sales – Gross cost of production
4. Operating Profit = Gross profit – Financial expenses – SLM depreciation
5. Taxable Profit = Gross profit – Financial expenses – WDV depreciation
6. Corporate Tax = 30% of Taxable profit
7. Profit after tax = Gross profit – Financial expenses – Corporate Tax
8. Profit for dividend = Profit after tax – SLM depreciation
9. Dividend = 25% of Profit for dividend
10. Profit after dividend = Profit for dividend – Dividend
11. Net Cash Accruals = Profit after dividend + SLM depreciation

Based on the projected working results, the cumulative cash flow for the project is represented in the form of a graph shown in **Error: Reference source not found**. It is assumed that project construction takes 2 years, and the cost of Land & Site Development is considered initially.

Payback period can be estimated as 2 years from the cumulative cash flow diagram.



₹ 10,000.00 Cr
₹ 8,000.00 Cr
₹ 6,000.00 Cr
₹ 4,000.00 Cr
₹ 2,000.00 Cr
-₹ 0.00
-₹ 2,000,000,000.00
-₹ 4,000,000,000.00
-4
-2
0
2
4
6
8
10
12
Cumulative Cash Flow

14.8: Discounted Profit Flow Analysis

This method considers the time value of profits and is used to determine the present value of the future profits and investments. The profit projections are discounted from day zero and prior investments are counted in the forward direction to determine their present value. This analysis involves the following steps:

5. Weighted Average Cost of Capital (WACC):

   WACC = (D * i_R + r_E * E) / (D + E)

   Where D is debt, E is equity, i_R is interest rate that is taken to be 12% and r_E is return on equity that is taken as 30%.

   Therefore, WACC = **21%**

6. Hurdle Rate of Return (K): The hurdle rate of return should be greater than WACC. Thus, K is taken as **25%**.
7. Present Value of Future Profits (P_V): The present value of profits is given by:

   $$P_V = \sum_{n=1}^{10} \frac{P_n}{(1+K)^n}$$

   Where, P_n = Gross profit projection for nth year, K = hurdle rate of return. Table 18.9 shows the Present Values of Future Profit. Summing up all the discounted profit values, we get P_v = ₹7,888.17 Cr
8. Present value of Investment (I_0): The total capital investment is **₹ 1,837.19 Cr.** We assume that the construction of the project takes 2 years, and 75% of the total investment is done in the first year itself. Thus, I_1 = **₹ 1,377.89 Cr** and I2 = **₹ 459.30 Cr**. Then, prevent value of investment = I_0 = (1 + K) * I_1 + I_2 = **₹ 2,181.66 Cr**
9. Net Present Value (NPV): The Net Present Value is given by: NPV = P_V – I_0 = **₹ 5,706.51 Cr** (which is >0)
10. Profitability Index = P_V / I_0 = **3.62** (which is >1)
11. Internal Rate of Return (IRR): IRR is estimated using: $I_0 = \sum_{n=1}^{10} \frac{P_n}{(1+IRR)^n}$

It is value of K for which the value of NPV is zero. Solving iteratively, **IRR = 101.2%.**

IRR > WACC

12. Economic Profit Analysis (EP):

Economic profit is given by,

EP = Gross Profit – Corporate Tax – SLM Depreciation – WACC * Project Cost – Interest on working capital * Borrowed working capital

Substituting a; the calculated values, we get EP = **₹ 1,131.93 Cr** (which is >0)

15. Conclusion

The preliminary techno-economic feasibility report to manufacture 23,000 tonnes per annum (TPA) of Methacrylic Acid (MAA) and 2-Hydroxyethyl methacrylate (HEMA) has been prepared. The desired products find extensive applications in the production of acrylic resins, surface coatings, adhesives, automotive paints, medical polymers, and dental materials. Thorough literature review has been conducted to select appropriate production processes: MAA is produced via two-stage catalytic oxidation of isobutene, while HEMA is produced via direct esterification of MAA with ethylene oxide. Both processes have been evaluated to be thermodynamically, kinetically, and economically feasible under the conditions prevalent in Dahej, Gujarat.

The selected production process for MAA involves the catalytic oxidation of isobutene with air in a fixed-bed reactor at elevated temperatures (350°C) to produce methacrolein, followed by a second-stage oxidation to methacrylic acid. The crude MAA is then purified through distillation and crystallization. For HEMA, the purified MAA is reacted with ethylene oxide in the presence of a catalyst and polymerization inhibitor to prevent unwanted side reactions. The reaction mass is then successively sent through various downstream separation and purification processes, including neutralization, solvent extraction, distillation, and filtration, to finally obtain high-purity HEMA, which is stored with added inhibitors to prevent polymerization during transportation. Suitable and justified assumptions have been made in the report in the absence of data to determine the required details.

A detailed economic analysis of the project has been done to determine whether the project is economically viable. The total project cost is estimated at ₹ 2,554.67 Cr, which acts as the initial investment required to set up the plant in Dahej, Gujarat. This sum is partially taken as a term loan (approximately ₹1,840 Crore) and partially as equity participation (approximately ₹920 Crore). The gross cost of production is calculated as ₹80,437.9 per tonne of product, whereas the working capital is estimated to be ₹1837.2 Crore, of which approximately ₹367.44 Crore is borrowed. Considering idealistic markets and full capacity utilization, the annual sales revenue is estimated to be ₹4764 Crore, resulting in a gross profit of ₹2,209 Crore per annum. The break-even capacity of the plant is approximately 7,682 TPA (32.25% of installed capacity), below which the project will generate negative profit. The payback period of the project is estimated at 2 years with a Return on Equity (ROE) of approximately 30%. The profitability index is calculated to be 3.62, which indicates that the project is profitable and can be considered for commissioning. The Net Present Value (NPV) over a 10-year project life, considering a discount rate of 21%, is approximately ₹5,700 Crore, which is greater than zero.

Therefore, the project is profitable and can attract investment with reasonable returns for a petrochemical venture of this scale. However, the assumptions made in this preliminary proposal must be kept in mind, and a more detailed analysis, including sensitivity analysis for raw material (isobutene and ethylene oxide) price fluctuations and product market demand, needs to be carried out before the actual commissioning of the project. Since the project relies heavily on the availability of isobutene (a petrochemical feedstock), locating the facility within the PCPIR in Dahej provides significant advantages in terms of feedstock accessibility and logistics. Profits may be further increased if the by-products (e.g., methacrolein recycle or acetic acid recovery) can be effectively recovered and sold, and if process heat integration opportunities are fully exploited to reduce utility costs. Additionally, the implementation of robust safety systems, given the flammability and toxicity of intermediates like isobutene and methacrolein, as well as the corrosive nature of MAA, is critical for both operational success and regulatory compliance.

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